Hydraulic pumps and oil injectors selection guide

SKF offers a wide range of hydraulic equipment, which facilitates the dismounting of bearings and other components. This selection guide features the most common applications for which the equipment can be used.

For more details on these hydraulic pumps and oil injectors, please see pages 29 - 36 of the Mounting and Lubrication section of this catalogue.

Ordering details and dimensons

Max. working pressure	Pump	Type	Oil container capacity	Dismounting applications*
30 MPa (4,350 psi)	**THAP 030**	Air–driven pump	Separate container oil	OK Couplings hydraulic chamber
50 MPa (7,250 psi)	**TMJL 50**	Hand operated pump	2 700 cm³ (165 in³)	≥ HMV 92E with sleeves OK Couplings
100 MPa (14,500 psi)	**729124**	Hand operated pump	250 cm³ (15 in³)	≤ HMV 54E with sleeves Oil injection for small bearings
	TMJL 100	Hand operated pump	800 cm³ (48 in³)	≤ HMV 92E with sleeves Oil injection for medium bearings
150 MPa (21,750 psi)	**THAP 150**	Air-driven pump	Separate container	Bolt tensioners, propellers Oil injection for bearing seatings
	728619 E	Hand operated pump	2 550 cm³ (155 in³)	All HMV E nuts with sleeves Oil injection for bearing seatings
300 MPa (43,500 psi)	**THAP 300E**	Air-driven pump	Separate container	OK Couplings Large pressure joints Oil injection for bearing seatings
	226400	Hand operated oil injector	200 cm³ (12,2 in³)	OK Couplings Adapter / withdrawal sleeves Oil injection for bearing seatings
	729101 B	Oil injection kit	200 cm³ (12,2 in³)	Complete kit / set to suit many applications
	TMJE 300	Oil injection set	200 cm³ (12,2 in³)	Complete kit / set to suit many applications
	226270	Screw injector	5,5 cm³ (0,33 in³)	Machine tool applications shaft diameter ≤ 100 mm
	226271	Screw injector	25 cm³ (1,5 in³)	Machine tool applications shaft diameter ≤ 200 mm
400 MPa (58,000 psi)	**THAP 400E**	Air-driven pump	Separate container	OK Couplings
	226400/ 400MPa	Hand operated oil injector	200 cm³ (12,2 in³)	Joints with high interference fits
	729101 E	Oil injection kit	200 cm³ (12,2 in³)	Complete kit / set to suit many applications
	TMJE 400	Oil injection set	200 cm³ (12,2 in³)	Complete kit / set to suit many applications

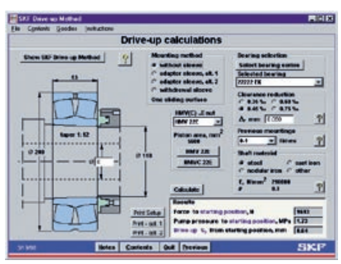
* *The dismounting applications given above are for guidance only.*
The interference fit present may mean that a pump / injector with a higher-pressure capacity is required.

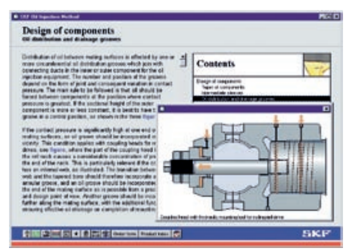
SKF Support

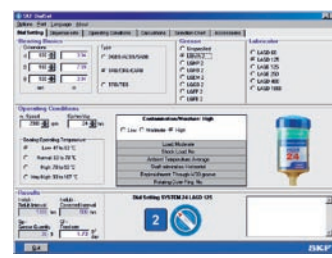
SKF Support

Reduced machinery downtime through effective bearing maintenance

Product quality is just one of the factors that determine bearing service life. Operating environment, proper installation and maintenance are also critical to bearing performance; factors that enter the picture after the bearing has been delivered to our customers.







SKF Internet sites

At www.mapro.skf.com you will find the SKF Maintenance Products catalogue online, offering a complete listing of products and technical specifications in many languages. There, you can also find a wealth of information about bearing maintenance practices as well as an extensive “frequently asked questions” section. For information on the SKF Group, history, products, divisions and services worldwide visit the SKF Group site at www.skf.com.

SKF Drive-up Method

The SKF Drive-up Method CD-ROM program is a computerized handbook on how to use the Drive-up Method for mounting bearings with a tapered bore. The program describes the method with the aid of pictures, animations, videos, and calculation tables that can easily be printed. The program is available in English, German, Swedish, French, Italian and Spanish. Reference no. MP3600

SKF Oil Injection Method

The SKF Oil Injection Method allows bearings and other components with an interference fit to be fitted and removed in a safe, controllable and rapid manner. The CD-ROM revolutionizes the method by fully automating the technique, making the detailed calculations easy and simple to compute. The CD-ROM provides detailed instructions and practical information on how to use the method for mounting and dismounting bearings, as well as using the method in design, calculation and application of shrink fitted components. Reference no. MP3601

SKF DialSet 4.0 Re-lubrication calculation program

The SKF DialSet Re-lubrication Calculation program enables accurate calculation of re-lubrication intervals for lubricating bearings. The computer program determines the right time setting and dispense rates for SYSTEM 24 and SYSTEM MultiPoint. The program is available on CD-ROM in English, French, German, Swedish, Spanish, Italian, Portuguese, Russian, Chinese and Thai languages. Reference no. MP3506. It is also available in English online as well as a downloadable version for PDA from www.mapro.skf.com.

At SKF we have put together the industry's most comprehensive program for maximising bearing service life and to help our customers reduce costly machine stoppages due to bearing failures..

For more information on the services described below, please contact your nearest SKF supplier

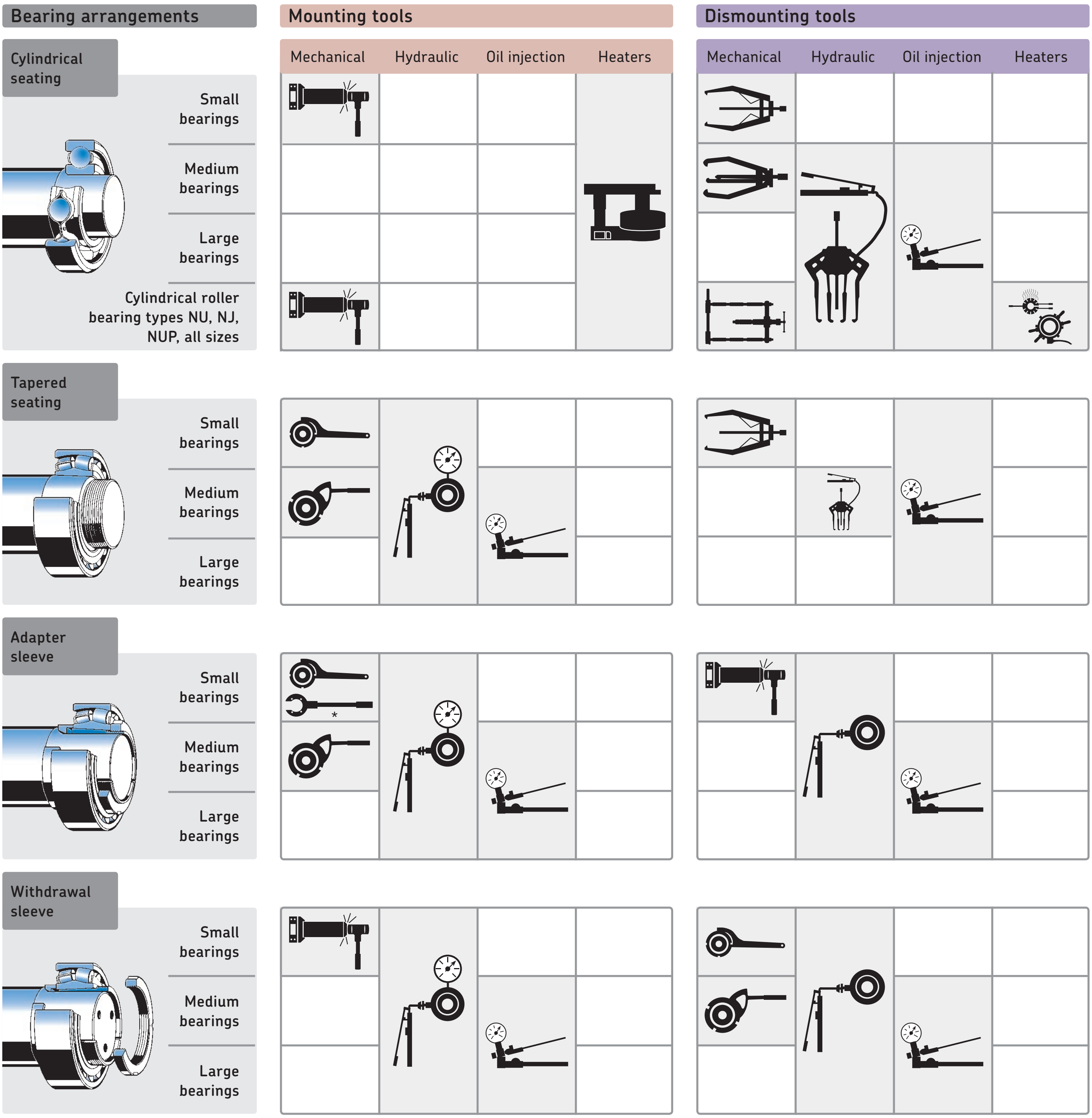
SKF Demonstration trucks
SKF offers demonstrations and training with mobile demonstration vehicles which tour throughout Europe, Asia and North America. The training program is tailored to the customers needs and may consist of a short theoretical explanation of the latest maintenance methods and concepts, followed by a detailed hands-on demonstration with qualified SKF personnel. For more information about these vehicles, please contact your local distributor or SKF office to schedule an appointment.

Audio visual
SKF provides a range of videos to support training courses on different facets of bearing and seal performance.
The "Get Even Smarter" video shows the do's and don't-s of good bearing maintenance in a light-hearted way.

Technical literature
SKF technical literature is a must in every maintenance workshop. The SKF General Catalogue and the unique SKF Bearing Maintenance Handbook provide the answers to all mounting and dismounting questions.

Training
SKF offers training courses on all facets of bearing maintenance and machine reliability. By prior arrangement training courses can be organised at our customers own premises or at one of our well-equipped SKF Maintenance Support Centres. If you would like to know what courses are offered in your area, talk to your local SKF representative or visit www.skf.com.

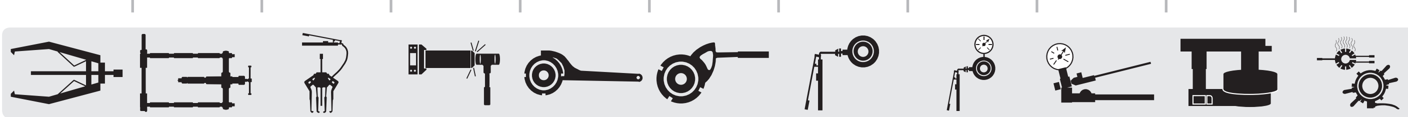
SKF methods and tools



Small bearings: bore diameter < 80 mm / Medium bearings: bore diameter 80 - 200 mm / Large bearings: bore diameter > 200 mm / * Only for self-aligning ball bearings.

Key

Jaw puller	Bearing separator	Hydraulic puller	Fitting tool	Hook spanner	Impact spanner	Hydraulic nut and pump	Drive-up Method	Oil Injection Method	Hot plate induction heater	Aluminium ring EAZ heater



TMFT 36 (page 11)

Designation	TMFT 36
Description	Fitting tool kit
Impact rings	Bore diameter 10 – 55 mm (0.39 – 2.1 in) Outer diameter: 26 – 120 mm (1.02 – 4.7 in)
Sleeves	Bore diameter: 18,5, 37,5 and 57,5 mm (0.7, 1.5, 2.3 in) Outer diameter: 25, 45 and 66 mm (1.0, 1.7, 2.6 in)
Hammer	36-H, weight 1 kg (2.2 lb)
Dimensions of the case	525 × 420 × 130 mm (20.6 × 16.5 × 5.1 in)
Number of rings	36
Number of sleeves	3
Weight (including carrying case)	4 kg (8.8 lb)

TMHN 7 (page 14)

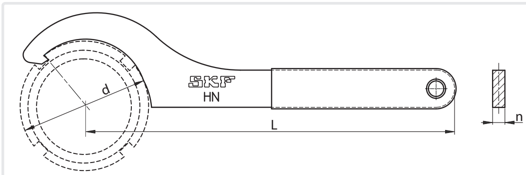
Designation	TMHN 7
Dimensions of case (w × d × h)	340 × 250 × 80 mm (13.4 × 9.8 × 3.1 in)
Weight	2,2 kg (4.7 lb)

TMHN 7 selection chart

Spanner	Bearing designation				Spanner	Bearing designation			
HNM 5	1205 EK	2205 EK	1305 EK		HNM 9	1209 EK	2209 EK	1309 EK	2309 EK
HNM 6	1206 EK	2206 EK	1306 EK	2306 K	HNM 10	1210 EK	2210 EK	1310 EK	2310 K
HNM 7	1207 EK	2207 EK	1307 EK	2307 EK	HNM 11	1211 EK	2211 EK	1311 EK	2311 K
HNM 8	1208 EK	2208 EK	1308 EK	2308 EK					

HN series (page 13)

Designation	HN ... (see table below)
Description	Hook spanner
Material	Special hardened steel
Material handle	PVC
Suitable	For many SKF nuts For all KM nuts according to DIN 981 For all nuts according to DIN 1804 For nuts from KM0 (Ø18) to KM22 (Ø145)



Designation	Spanner design DIN 1810	Diameter d		Working length L		Thickness n		Weight	
	mm	mm	in	mm	in	mm	in	g	lb
HN 0		16 – 20	0.6 – 0.8	100	3.9	3	0.12	24	0.05
HN 1	Ø20 – Ø22	20 – 22	0.8 – 0.9	100	3.9	3	0.12	25	0.06
HN 2-3	Ø25 – Ø28	25 – 28	1.0 – 1.1	120	4.7	4	0.16	48	0.11
HN 4	Ø30 – Ø32	30 – 32	1.2 – 1.3	120	4.7	4	0.16	48	0.11
HN 5-6		38 – 45	1.5 – 1.8	150	5.9	5	0.20	96	0.21
HN 7	Ø52 – Ø55	52 – 55	2.0 – 2.2	180	7.1	6	0.24	170	0.37
HN 8-9		58 – 65	2.3 – 2.6	210	8.3	7	0.28	270	0.60
HN 10-11	Ø68 – Ø75	68 – 75	2.7 – 3.0	210	8.3	7	0.28	270	0.60
HN 12-13	Ø80 – Ø90	80 – 90	3.1 – 3.5	240	9.4	8	0.31	420	0.93
HN 14		92	3.6	240	9.4	8	0.31	415	0.91
HN 15	Ø95 – Ø100	95 – 100	3.7 – 3.9	240	9.4	8	0.31	405	0.89
HN 16		105	4.1	240	9.4	8	0.31	412	0.91
HN 17	Ø110 – Ø115	110 – 115	4.3 – 4.5	280	11.0	10	0.39	753	1.66
HN 18-20	Ø120 – Ø130	120 – 130	4.7 – 5.1	280	11.0	10	0.39	752	1.66
HN 21-22	Ø135 – Ø145	135 – 145	5.3 – 5.7	320	12.6	12	0.47	1210	2.67

HN series selection chart

	Suitable for SKF nuts of series						
	KM	N	AN	KMK	KMFE	KMT	DIN 1804 (M)
HN 0	0	0		0			M6 × 0,75, M8 × 1
HN 1	1	1		1			M8 × 1
HN 2-3	2, 3	2, 3		2, 3		0	M10 × 1, M12 × 1,5
HN 4	4	4		4	4	1, 2	M14 × 1,5, M16 × 1,5
HN 5-6	5, 6	5, 6		5, 6	5, 6	3, 4, 5	M22 × 1,5, M24 × 1,5, M26 × 1,5
HN 7	7	7		7	7	6, 7	M32 × 1,5, M35 × 1,5
HN 8-9	8, 9	8, 9		8, 9	8, 9	8	M38 × 1,5, M40 × 1,5, M42 × 1,5
HN 10-11	10, 11	10, 11		10, 11	10, 11	9, 10	M45 × 1,5, M48 × 1,5, M50 × 1,5
HN 12-13	12, 13	12, 13		12, 13	12, 13	11, 12	M52 × 1,5, M55 × 1,5, M58 × 1,5, M60 × 1,5
HN 14	14		14	14	14		
HN 15	15		15	15	15	13, 14	M62 × 1,5, M65 × 1,5, M68 × 1,5, M70 × 1,5
HN 16	16		16	16	16	15	
HN 17	17		17	17	17	16	M72 × 1,5, M75 × 1,5, M80 × 2
HN 18-20	18, 19, 20		18, 19, 20	18, 19, 20	18, 19, 20	17, 18, 19	M85 × 2, M90 × 2
HN 21-22	21, 22		21, 22		21, 22	20, 22	M95 × 2, M100 × 2

Technical data

HN 4-16/SET (page 13)

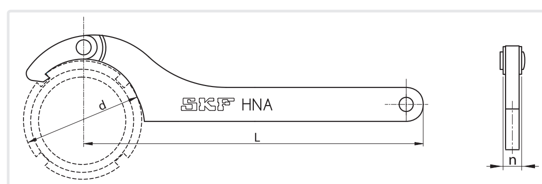
Designation	HN 4-16/SET		
Description	Hook spanner set	Suitable	For many SKF nuts
Material	Special hardened steel		For KM nuts according to DIN 981
Material handle	PVC		For nuts according to DIN 1804
Dimensions	645 x 320 mm (25.4 x 12.6 in)		For nuts from KM4 to KM16
Weight	2,7 kg (6 lb)		

Designation	DIN 1810	Diameter		Working length		Thickness		Weight	
	mm	mm	in	mm	in	mm	in	g	lb
HN 4	Ø30 - Ø32	30 - 32	1.2 - 1.3	120	4.7	4	0.16	48	0.11
HN 5-6		38 - 45	1.5 - 1.8	150	5.9	5	0.20	96	0.21
HN 7	Ø52 - Ø55	52 - 55	2.0 - 2.2	180	7.1	6	0.24	170	0.37
HN 8-9		58 - 65	2.3 - 2.6	210	8.3	7	0.28	270	0.60
HN 10-11	Ø68 - Ø75	68 - 75	2.7 - 3	210	8.3	7	0.28	270	0.60
HN 12-13	Ø80 - Ø90	80 - 90	3.1 - 3.5	240	9.4	8	0.31	420	0.93
HN 14		92	3.6	240	9.4	8	0.31	415	0.91
HN 15	Ø95 - Ø100	95 -100	3.7 - 3.9	240	9.4	8	0.31	405	0.89
HN 16		105	4.1	240	9.4	8	0.31	412	0.91

HNA series (page 13)

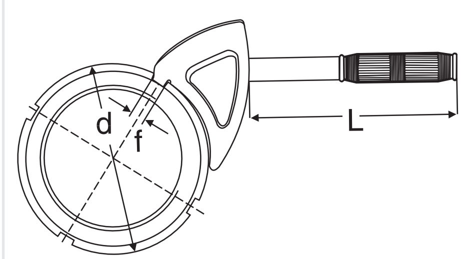
Designation	Description	Diameter d		Working length L		Thickness n		Weight	
		mm	in	mm	in	mm	in	g	lb
HNA 1-4	size 2 – 4	20 – 35	0.8 – 1.4	145	5.7	6	0.24	50	0.11
HNA 5-8	size 5 – 8	35 – 60	1.4 – 2.4	180	7.1	8	0.31	100	0.22
HNA 9-13	size 9 – 13	60 – 90	2.4 – 3.5	240	9.5	10	0.39	285	0.63
HNA 14-24	size 14 – 24	90 – 15	3.5 – 6.1	275	10.8	12	0.47	450	0.99

Designation	Suitable for SKF nuts of series						
	KM	KML	N	AN	KMK	KMFE	KMT
HNA 1-4	2 – 4		2 – 4		2 – 4	4	0 – 2
HNA 5-8	5 – 8		5 – 8		5 – 8	5 – 8	3 – 7
HNA 9-13	9 – 13		9 – 13		9 – 13	9 – 13	8 – 12
HNA 14-24	14 – 24	24		14 – 24	14 – 20	14 – 24	13 – 24



TMFN series (page 13)

Designation	Dimensions d		f		L		Weight	
	mm	in	mm	in	mm	in	kg	lb
TMFN 23-30	150 – 195	5.9 – 7.7	11,5	0.45	200	7.9	1,1	2.4
TMFN 30-40	195 – 250	7.7 – 9.8	13,5	0.53	200	7.9	1,5	3.3
TMFN 40-52	250 – 320	9.8 – 12.6	17	0.67	340	13.4	3,2	7.0
TMFN 52-64	320 – 400	12.6 – 15.7	19	0.75	325	12.8	4,1	9.0
TMFN 64-80	400 – 520	15.7 – 20.5	23	0.91	310	12.2	4,3	9.5
TMFN 80-500	520 – 630	20.5 – 24.8	28	1.10	370	14.6	6,9	15.2
TMFN 500-600	630 – 750	24.8 – 29.5	36	1.42	350	13.8	8,5	18.7
TMFN 600-750	750 – 950	29.5 – 37.4	40	1.57	600	23.6	11,0	24.2

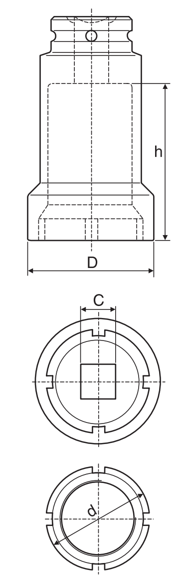


TMFN series selection chart

Designation	Suitable for adapter sleeves (sizes) H 23, H 31 H 32	H 30 H 39	Suitable for nuts of series KM	KML	HM T	HM	KMFE	KMT	DIN 1804 (M)
TMFN 23-30	24 – 30	26 – 32	23 – 30	26 – 32	–	–	23 – 26	24, 26–32	M105 × 2, M110 × 2
TMFN 30-40	30 – 40	34 – 40	31 – 40	34 – 40	–	–	–	34 – 40	–
TMFN 40-52	40 – 48	44 – 52	–	–	42T – 50T	3044 – 3052	–	–	–
TMFN 52-64	52 – 64	56 – 68	–	–	52T – 56T	3056 – 3068	–	–	–
TMFN 64-80	64 – 80	68 – 88	–	–	–	3168 – 3088	–	–	–
TMFN 80-500	80 – 500	88 – 530	–	–	–	3184 – 3196	–	–	–
TMFN 500-600	500 – 600	530 – 630	–	–	–	30/500 – 30/630	–	–	–
TMFN 600-750	600 – 750	670 – 800	–	–	–	31/600 – 31/750	–	–	–

TMFS series (page 15)

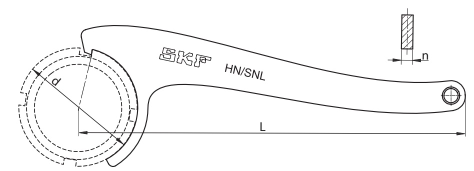
Designation	Dimensions d mm	d inch	D mm	D inch	h mm	h inch	Connection □ C inch	Weight kg	Weight lb	Suitable for nuts of series KM, KMK, KMF size
TMFS 0	18	0.7	22,0	0.9	45	1.8	3/8	0,12	0.27	0
TMFS 1	22	0.9	28,0	1.1	45	1.8	3/8	0,12	0.27	1
TMFS 2	25	1.0	33,0	1.3	61	2.4	1/2	0,22	0.49	2
TMFS 3	28	1.1	36,0	1.4	61	2.4	1/2	0,23	0.51	3
TMFS 4	32	1.3	38,0	1.5	58	2.3	1/2	0,26	0.58	4
TMFS 5	38	1.5	46,0	1.8	58	2.3	1/2	0,34	0.75	5
TMFS 6	45	1.8	53,0	2.1	58	2.3	1/2	0,39	0.86	6
TMFS 7	52	2.0	60,0	2.4	58	2.3	1/2	0,45	1.00	7
TMFS 8	58	2.3	68,0	2.7	58	2.3	1/2	0,51	1.13	8
TMFS 9	65	2.6	73,5	2.9	63	2.5	3/4	0,89	1.97	9
TMFS 10	70	2.8	78,5	3.1	63	2.5	3/4	0,79	1.75	10
TMFS 11	75	3.0	83,5	3.3	63	2.5	3/4	0,87	1.92	11
TMFS 12	80	3.1	88,5	3.5	63	2.5	3/4	1,40	3.09	12
TMFS 13	85	3.3	94,0	3.7	63	2.5	3/4	1,40	3.09	13
TMFS 14	92	3.6	103,0	4.1	80	3.2	1	1,92	4.24	14
TMFS 15	98	3.9	109,0	4.3	80	3.2	1	1,92	4.02	15
TMFS 16	105	4.1	116,0	4.6	80	3.2	1	1,83	4.04	16
TMFS 17	110	4.3	121,0	4.8	80	3.2	1	1,83	4.04	17
TMFS 18	120	4.7	131,0	5.2	80	3.2	1	3,60	7.94	18
TMFS 19	125	4.9	137,0	5.5	80	3.2	1	3,05	6.73	19
TMFS 20	130	5.1	143,0	5.7	80	3.2	1	3,30	7.28	20



HN /SNL series (page 14)

Designation HN **/SNL

Description Special hook spanner for use with SNL housings
Material Black phosphated, hardened chrome vanadium steel
Suitable SKF SNL and SNH housings
KM, KML, N, AN, KMK, KMFE and KMT lock nuts



Designation	d – outer diameter locknut mm	in	L – working length mm	in	n – thickness mm	in	weight g	lb
HN 5/SNL	38	1.50	175	6.9	5	0.20	100	0.22
HN 6/SNL	45	1.77	210	8.3	6	0.24	176	0.39
HN 7/SNL	52	2.05	210	8.3	6	0.24	180	0.40
HN 8/SNL	58	2.28	245	9.6	7	0.28	280	0,62
HN 9/SNL	65	2.56	245	9.6	7	0.28	295	0.65
HN 10/SNL	70	2.76	245	9.6	7	0.28	310	0.68
HN 11/SNL	75	2.95	245	9.6	7	0.28	330	0.73
HN 12/SNL	80	3.15	280	11.0	8	0.31	455	1.00
HN 13/SNL	85	3.35	280	11.0	8	0.31	484	1.07
HN 15/SNL	98	3.86	280	11.0	8	0.31	490	1.08
HN 16/SNL	105	4.13	325	12.8	10	0.39	780	1.72
HN 17/SNL	110	4.33	325	12.8	10	0.39	826	1.82
HN 18/SNL	120	4.72	325	12.8	10	0.39	826	1.82
HN 19/SNL	125	4.92	325	12.8	10	0.39	865	1.91
HN 20/SNL	130	5.12	325	12.8	10	0.39	875	1.93
HN 22/SNL	145	5.71	375	14.8	12	0.47	1260	2.78
HN 24/SNL	155	6.10	375	14.8	12	0.47	1352	2.98
HN 26/SNL	165	6.50	375	14.8	12	0.47	1395	3.08
HN 28/SNL	180	7.09	445	17.5	14	0.55	2175	4.80
HN 30/SNL	195	7.68	445	17.5	14	0.55	2281	5.03
HN 32/SNL	210	8.27	445	17.5	14	0.55	2486	5.48

HN /SNL series selection chart

	Suitable for SKF housings SNL	Suitable for SKF nuts of series KM	KML	N	AN	KMK	KMFE	KMT*
HN 5/SNL	505, 506 – 605	5		5		5	5	5
HN 6/SNL	506 – 605, 507 – 606	6		6		6	6	6
HN 7/SNL	507 – 606, 508 – 607	7		7		7	7	7
HN 8/SNL	508 – 607, 510 – 608	8		8		8	8	
HN 9/SNL	509, 511 – 609	9		9		9	9	8
HN 10/SNL	510 – 608, 512 – 610	10		10		10	10	9
HN 11/SNL	511 – 609, 513 – 611	11		11		11	11	10
HN 12/SNL	512 – 610, 515 – 612	12		12		12	12	
HN 13/SNL	513 – 611, 516 – 613	13		13		13	13	11, 12, 13
HN 15/SNL	515 – 612, 518 – 615	15			15	15	15	14
HN 16/SNL	516 – 613, 519 – 616	16			16	16	16	15
HN 17/SNL	517, 520 – 617	17			17	17	17	16

Technical data

HN /SNL series selection chart

	Suitable for SKF housings	Suitable for SKF nuts of series						
	SNL	KM	KML	N	AN	KMK	KMFE	KMT*
HN 18/SNL	518 – 615	18			18	18	18	17
HN 19/SNL	519 – 616, 522 – 619	19			19	19	19	18
HN 20/SNL	520 – 617, 524 – 620	20			20	20	20	19, 20
HN 22/SNL	522 – 619	22	24		22		22	22
HN 24/SNL	524 – 620	24	26		24		24	24
HN 26/SNL	526	26	28				26	26, 28
HN 28/SNL	528	28	30		28			30
HN 30/SNL	530	30	32		30			32
HN 32/SNL	532	32						34

** Not recommended in combination with SNL/SNH housing*

TMBH 1 (page 17)

Designation	**TMBH 1**
Power:	
Voltage	100 – 240 V, 50 – 60 Hz
Power (maximum)	350 Watt
Cosine φ	> 0,95
Component size range:	
– inner diameter	20 ... 100 mm (0.8 ... 4 in)
– width	< 50 mm (2 in)
– weight	up to approximately 5 kg (11 lb)
Control functions:	
Time control	0 – 60 minutes
Temperature control	0 – 200 °C (32 – 392 °F)
Accuracy temperature control	± 3 °C (6 °F)
Maximum temperature	200 °C (392 °F)
Dimensions:	
Control box	150 × 330 × 105 mm (6 × 13 × 4 in)
Heating clamp	114 × 114 mm (4.5 × 4.5 in)
Operating space heating clamp	52 × 52 mm (2 × 2 in)
Complete unit in carrying case	370 × 240 × 130 mm (15 × 9 × 5 in)
Length clamp cable	75 cm (30 in)
Length power cable	2 m (80 in)
Length temperature probe cable	100 cm (40 in)
Weight complete unit	4,5 kg (10 lb)

729659 C (page 17)

Designation	**729659 C** **729659 C/110V**
Voltage	729659 C 230V (50/60Hz) 729659 C/110V 115V (50/60Hz)
Power	1 000 W
Temperature range	50 – 200 °C (120 – 390 °F)
Plate dimensions (l × w)	380 × 178 mm (15 × 7 in)
Height of cover	50 mm (2 in)
Overall dimensions (l × w × h)	400 × 240 × 130 mm (16 × 10 × 5 in)
Weight	4,7 kg (10 lb)
Length of connection cable	2 metres (6.6 ft) (earth connection required)

TIH ...m series (page 18-20)

Designation	TIH 030M	TIH 100M	TIH 220M
SKF m20 performance	28 kg (61.7 lb)	97 kg (213 lb)	220 kg (480 lb)
Voltage, V/Hz	230 V/50–60 Hz or 110 V/50–60 Hz	230 V/50–60 Hz or 400–460 V/50–60 Hz	200–230/50–60 Hz or 400–460/50–60 Hz
Work piece (bearings): – Maximum weight – Maximum bore diameter	 40 kg (88 lb) 20 – 300 mm (0.8 – 11.8 in)	 120 kg (264 lb) 20 – 400 mm (0.8 – 15.7 in)	 300 kg (661,4 lb) 60 – 600 mm (2.3 – 23.6 in)
Temperature control: – Range – Magnetic probe – Accuracy (electronics)	 0 – 250 °C (32 – 482 °F) Yes, K-type ± 2 °C (± 3.6 °F)	 0 – 250 °C (32 – 482 °F) Yes, K-type ± 2 °C (± 3.6 °F)	 0 – 250 °C (32 – 482 °F) Yes, K-type ± 2 °C (± 3.6 °F)
Time control: – Range – Accuracy	 0 – 60 minutes ± 0,01 sec.	 0 – 60 minutes ± 0,01 sec.	 0 – 60 minutes ± 0,01 sec.
Maximum temperature (approx.)	400 °C (750 °F)	400 °C (750 °F)	400 °C (750 °F)
Thermometer mode	Yes	Yes	Yes
Bearing mode (pre-set at 110 °C / 230 °F)	Yes	Yes	Yes
Power reduction	2-step; 50 – 100%	2-step; 50 – 100%	2-step; 50 – 100%
Demagnetisation according to SKF norms (automatic)	Yes (<2 A/cm)	Yes (<2 A/cm)	Yes (<2 A/cm)
Can heat sealed bearings	Yes	Yes	Yes
Can heat pre-greased bearings	Yes	Yes	Yes
Error guiding codes	Yes	Yes	Yes
Thermal overload protection	Yes	Yes	Yes
Maximum magnetic flux	1,7 T	1,7 T	1,55 T
Control panel	Key board with LED in remote control	Key board with LED in remote control	Key board with LED in remote control
Operating area (w × h)	100 × 135 mm (3.9 × 5.3 in)	155 × 205 mm (6.1 × 8.0 in)	250 × 255 mm (9.8 × 10 in)
Coil diameter	95 mm (3.7 in)	110 mm (4.3 in)	140 mm (5.5 in)
Dimensions (w × d × h)	450 × 195 × 210 mm (17.7 × 7.6 × 8.2 in)	570 × 230 × 350 mm (22.4 × 9.0 × 13.7 in)	750 × 290 × 440 mm (29.5 × 11.4 × 17.3 in)
Total weight, including yokes	20,9 kg (46 lb)	42 kg (92 lb)	86 kg (189 lb)
Maximum power consumption	2,0 kVA	3,6 kVA (230 V) 4,0-4,6 kVA (400-460 V)	10,0-11,5 kVA (400-460 V)
Number of standard yokes	3	3	2
Standard yokes	45 × 45 × 215 mm (1.7 × 1.7 × 8.4 in), for heating bearings with bore diameter of 65 mm (2,6 in) and larger 28 × 28 × 215 mm (1.1 × 1.1 × 8.4 in), for heating bearings with bore diameter of 40 mm (1.6 in) and larger 14 × 14 × 215 mm (0.5 × 0.5 × 8.4 in), for heating bearings with bore diameter of 20 mm (0.8 in) and larger	56 × 56 × 296 mm (2.2 × 2.2 × 11.7 in), for heating bearings with bore diameter of 80 mm (3.1 in) and larger 28 × 28 × 296 mm (1.1 × 1.1 × 11.7 in), for heating bearings with bore diameter of 40 mm (1.6 in) and larger 14 × 14 × 296 mm (0.6 × 0.6 × 11.7 in), for heating bearings with bore, diameter of 20 mm (0.8 in) and larger	70 × 70 × 430 mm (2.8 × 2.8 × 16.9 in), for heating bearings with bore diameter of 100 mm (3.9 in) and larger 40 × 40 × 430 mm (1.6 × 1.6 × 16.9 in), for heating bearings with bore diameter of 60 mm (2.4 in) and larger
Core cross section	45 × 45 mm (1.7 × 1.7 in)	56 × 56 mm (2.2 × 2.2 in)	70 × 70 mm (2.8 × 2.8 in)
Yoke storage	Yes, foldable	Yes, foldable	Yes, foldable
Sliding arm	No	No	Yes, 70 × 70 × 430 mm (2.8 × 2.8 × 16.9 in) yoke only
Swivel arm	No	Yes, 56 × 56 × 296 mm (2.2 × 2.2 × 11.7 in) yoke only	No
Cooling fan	No	No	No
Housing material	Steel and glass filled polyamide	Steel and glass filled polyamide	Steel and glass filled polyamide
Warranty period	3 years	3 years	3 years

Technical data

TMMH series (page 15)

Designation	TMMH 300/500	TMMH 500/700
Bearing outer diameter D	300 – 500 mm (12 – 20 in)	500 – 700 mm (20 – 28 in)
Max. lifting weight	500 kg (1 100 lb)	500 kg (1 100 lb)
Weight	6,3 kg (14 lb)	6,3 kg (14 lb)

TIH T1 (page 21)

Designation	TIH T1		
Width	50 cm (20 in)	**Length**	72 cm (28 in)
Height	74 cm (29 in)	**Capacity**	900 kg (1 934 lb)

Drive-up Method: 729124 SRB, TMJL 100SRB and TMJL 50SRB (page 24)

Designation	729124 SRB	TMJL 100SRB	TMJL 50SRB
Max. pressure	100 MPa / 14 500 psi	100 MPa / 14 500 psi	50 MPa / 7 250 psi
Volume/stroke	0,5 cm^3 / 0.03 in^3	1,0 cm^3 / 0.06 in^3	3,5 cm^3 / 0.21 in^3
Oil container capacity	250 cm^3 / 15 in^3	800 cm^3 / 48 in^3	2 700 cm^3 / 165 in^3
Digital pressure gauge unit	MPa / psi	MPa / psi	MPa / psi

NOTE: All above pumps are complete with digital pressure gauge, high pressure hose and quick connect coupling.

Ordering details

Designation	Description	Designation	Description
HMV ..E (e.g. HMV 54E)	Metric thread hydraulic nut	**TMJG 100 D**	Gauge only (MPa/psi)
HMVC ..E (e.g. HMVC 54E)	Inch thread hydraulic nut	**TMCD 10R**	Horizontal dial indicator (0 – 10 mm)
HMV ..E/A101 (e.g. HMV 54E/A101)	Unthreaded hydraulic nut	**TMCD 5P**	Vertical dial indicator (0 – 5 mm)
729124 SRB (for nuts ≤ HMV 54E)	Pump with digital gauge (MPa/psi)	**TMCD 1/2R**	Horizontal dial indicator (0 – 0.5 in)
TMJL 100SRB (for nuts ≤ HMV 92E)	Pump with digital gauge (MPa/psi)		
TMJL 50SRB (all sizes HMV ..E nuts)	Pump with digital gauge (MPa/psi)		

HMV E series (page 26 and 119)

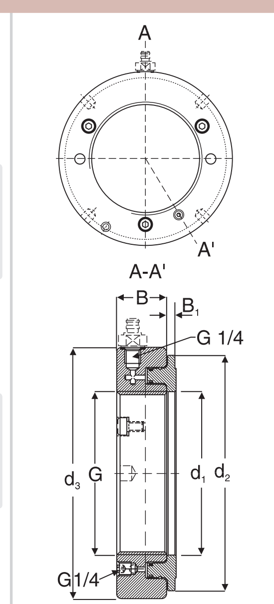
Designation	HMV E		
Thread form		**Recommended Pumps**	
HMV 10E – HMV 40E	ISO 965/111-1980 tolerance class 6H	**HMV 10E – HMV 54E**	729124 / TMJL 100 / 728619 E / TMJL 50
HMV 41E – HMV 200E	ISO 2901-1977 tolerance class 7H	**HMV 56E – HMV 92E**	TMJL 100 / 728619 E/ TMJL 50
Mounting fluid	LHMF 300	**HMV 94E – HMV 200E**	728619 E/ TMJL 50
		Quick connection nipple	729832 A (included)

Replacement parts			
O–rings	Nut designation followed by /233983 e.g. HMV 10/233983	**Other types available**	
		Inch series nuts	HMVC E series
Ball plug	233950E	**Nuts without threads**	HMV...E/A101
Quick connection nipple	729832 A		

Special executions also available on request

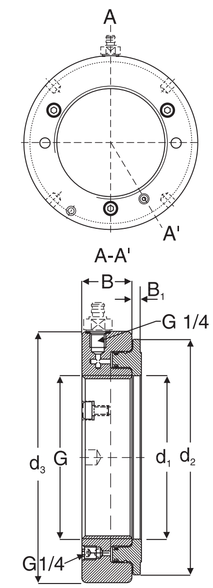
Ordering details and dimensions

Designation	G thread	d_1 mm	d_2 mm	d_3 mm	B mm	B_1 mm	Permitted piston displacement mm	Piston area mm^2	Weight kg
HMV 10E	M 50 × 1,5	50,5	104	114	38	4	5	2 900	2,70
HMV 11E	M 55 × 2	55,5	109	120	38	4	5	3 150	2,75
HMV 12E	M 60 × 2	60,5	115	125	38	5	5	3 300	2,80
HMV 13E	M 65 × 2	65,5	121	130	38	5	5	3 600	3,00
HMV 14E	M 70 × 2	70,5	127	135	38	5	5	3 800	3,20
HMV 15E	M 75 × 2	75,5	132	140	38	5	5	4 000	3,40
HMV 16E	M 80 × 2	80,5	137	146	38	5	5	4 200	3,70
HMV 17E	M 85 × 2	85,5	142	150	38	5	5	4 400	3,75
HMV 18E	M 90 × 2	90,5	147	156	38	5	5	4 700	4,00
HMV 19E	M 95 × 2	95,5	153	162	38	5	5	4 900	4,30
HMV 20E	M 100 × 2	100,5	158	166	38	6	5	5 100	4,40
HMV 21E	M 105 × 2	105,5	163	172	38	6	5	5 300	4,65
HMV 22E	M 110 × 2	110,5	169	178	38	6	5	5 600	4,95
HMV 23E	M 115 × 2	115,5	174	182	38	6	5	5 800	5,00
HMV 24E	M 120 × 2	120,5	179	188	38	6	5	6 000	5,25
HMV 25E	M 125 × 2	125,5	184	192	38	6	5	6 200	5,35
HMV 26E	M 130 × 2	130,5	190	198	38	6	5	6 400	5,65
HMV 27E	M 135 × 2	135,5	195	204	38	6	5	6 600	5,90
HMV 28E	M 140 × 2	140,5	200	208	38	7	5	6 800	6,00
HMV 29E	M 145 × 2	145,5	206	214	39	7	5	7 300	6,50



Ordering details and dimensions

Designation	G	d_1	d_2	d_3	B	B_1	Permitted piston displacement	Piston area	Weight
	thread	mm	mm	mm	mm	mm	mm	mm²	kg
HMV 30E	M 150 × 2	150,5	211	220	39	7	5	7 500	6,60
HMV 31E	M 155 × 3	155,5	218	226	39	7	5	8 100	6,95
HMV 32E	M 160 × 3	160,5	224	232	40	7	6	8 600	7,60
HMV 33E	M 165 × 3	165,5	229	238	40	7	6	8 900	7,90
HMV 34E	M 170 × 3	170,5	235	244	41	7	6	9 400	8,40
HMV 36E	M 180 × 3	180,5	247	256	41	7	6	10 300	9,15
HMV 38E	M 190 × 3	191	259	270	42	8	7	11 500	10,5
HMV 40E	M 200 × 3	201	271	282	43	8	8	12 500	11,5
HMV 41E	Tr 205 × 4	207	276	288	43	8	8	12 800	12,0
HMV 42E	Tr 210 × 4	212	282	294	44	8	9	13 400	12,5
HMV 43E	Tr 215×4	217	287	300	44	8	9	13 700	13,0
HMV 44E	Tr 220×4	222	293	306	44	8	9	14 400	13,5
HMV 45E	Tr 225×4	227	300	312	45	8	9	15 200	14,5
HMV 46E	Tr 230×4	232	305	318	45	8	9	15 500	14,5
HMV 47E	Tr 235×4	237	311	326	46	8	10	16 200	16,0
HMV 48E	Tr 240×4	242	316	330	46	9	10	16 500	16,0
HMV 50E	Tr 250×4	252	329	342	46	9	10	17 600	17,5
HMV 52E	Tr 260×4	262	341	356	47	9	11	18 800	19,0
HMV 54E	Tr 270×4	272	352	368	48	9	12	19 800	20,5
HMV 56E	Tr 280×4	282	363	380	49	9	12	21 100	22,0
HMV 58E	Tr 290 × 4	292	375	390	49	9	13	22 400	22,5
HMV 60E	Tr 300 × 4	302	386	404	51	10	14	23 600	25,5
HMV 62E	Tr 310 × 5	312	397	416	52	10	14	24 900	27,0
HMV 64E	Tr 320 × 5	322	409	428	53	10	14	26 300	29,5
HMV 66E	Tr 330 × 5	332	419	438	53	10	14	27 000	30,0
HMV 68E	Tr 340 × 5	342	430	450	54	10	14	28 400	31,5
HMV 69E	Tr 345 × 5	347	436	456	54	10	14	29 400	32,5
HMV 70E	Tr 350 × 5	352	442	464	56	10	14	29 900	35,0
HMV 72E	Tr 360 × 5	362	455	472	56	10	15	31 300	35,5
HMV 73E	Tr 365 × 5	367	460	482	57	11	15	31 700	38,5
HMV 74E	Tr 370 × 5	372	466	486	57	11	16	32 800	39,0
HMV 76E	Tr 380 × 5	382	476	498	58	11	16	33 500	40,5
HMV 77E	Tr 385 × 5	387	483	504	58	11	16	34 700	41,0
HMV 80E	Tr 400 × 5	402	499	522	60	11	17	36 700	45,5
HMV 82E	Tr 410 × 5	412	510	534	61	11	17	38 300	48,0
HMV 84E	Tr 420 × 5	422	522	546	61	11	17	40 000	50,0
HMV 86E	Tr 430 × 5	432	532	556	62	11	17	40 800	52,5
HMV 88E	Tr 440 × 5	442	543	566	62	12	17	42 500	54,0
HMV 90E	Tr 450 × 5	452	554	580	64	12	17	44 100	57,5
HMV 92E	Tr 460 × 5	462	565	590	64	12	17	45 100	60,0
HMV 94E	Tr 470 × 5	472	576	602	65	12	18	46 900	62,0
HMV 96E	Tr 480 × 5	482	587	612	65	12	19	48 600	63,0
HMV 98E	Tr 490 × 5	492	597	624	66	12	19	49 500	66,0
HMV 100E	Tr 500 × 5	502	609	636	67	12	19	51 500	70,0
HMV 102E	Tr 510 × 6	512	624	648	68	12	20	53 300	74,0
HMV 104E	Tr 520 × 6	522	634	658	68	13	20	54 300	75,0
HMV 106E	Tr 530 × 6	532	645	670	69	13	21	56 200	79,0
HMV 108E	Tr 540 × 6	542	657	682	69	13	21	58 200	81,0
HMV 110E	Tr 550 × 6	552	667	693	70	13	21	59 200	84,0
HMV 112E	Tr 560 × 6	562	678	704	71	13	22	61 200	88,0
HMV 114E	Tr 570 × 6	572	689	716	72	13	23	63 200	91,0
HMV 116E	Tr 580 × 6	582	699	726	72	13	23	64 200	94,0
HMV 120E	Tr 600 × 6	602	721	748	73	13	23	67 300	100
HMV 126E	Tr 630 × 6	632	754	782	74	14	23	72 900	110
HMV 130E	Tr 650 × 6	652	775	804	75	14	23	76 200	115
HMV 134E	Tr 670 × 6	672	796	826	76	14	24	79 500	120
HMV 138E	Tr 690 × 6	692	819	848	77	14	25	84 200	127
HMV 142E	Tr 710 × 7	712	840	870	78	15	25	87 700	135
HMV 150E	Tr 750 × 7	752	883	912	79	15	25	95 200	146
HMV 160E	Tr 800 × 7	802	936	965	80	16	25	103 900	161
HMV 170E	Tr 850 × 7	852	990	1 020	83	16	26	114 600	181
HMV 180E	Tr 900 × 7	902	1 043	1 075	86	17	30	124 100	205
HMV 190E	Tr 950 × 8	952	1 097	1 126	86	17	30	135 700	218
HMV 200E	Tr 1000 × 8	1 002	1 150	1 180	88	17	34	145 800	239



729124 (page 29)

Designation	729124		
Maximum pressure	100 MPa (14 500 psi)	Length of pressure hose	1 500 mm (59 in)
Volume/stroke	0,5 cm³ (0.03 in³)	Connection nipple	G 1/4 quick connection
Oil container capacity	250 cm³ (15 in³)	Weight	3,5 kg (8 lb)

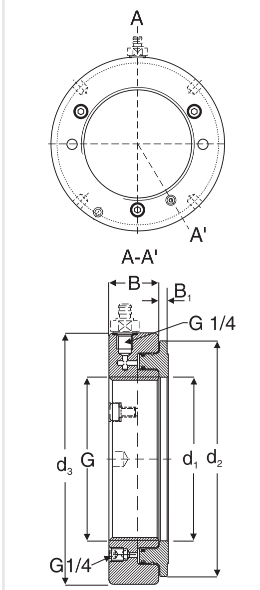
Technical data

HMVC E series (page 26 and 109)

Designation	HMVC E		
Thread form		**Recommended Pumps**	
HMVC 10E – HMVC 64E	American National Form Threads Class 3	**HMVC 10E – HMVC 52E**	729124 / TMJL 100 / 728619 E / TMJL 50
HMVC 68E – HMVC 190E	ACME General Purpose Threads Class 3 G	**HMVC 56E – HMVC 92E**	TMJL 100 / 728619 E / TMJL 50
Mounting fluid	LHMF 300	**HMVC 94E – HMVC 190E**	728619 E/ TMJL 50
		Quick connection nipple	729832 A (included)

Ordering details and dimensions

Designation	G	Pitch diameter	Threads per in	d_1	d_2	d_3	B	B_1	Permitted piston displacement	Piston area	Weight
	in	in	–	in	in	in	in	in	in	in²	lb
HMVC 10E	1 967	1 9309	18	2.0	4.1	4.5	1.5	0.16	0.20	4.5	6.0
HMVC 11E	2 157	2 1209	18	2.2	4.3	4.7	1.5	0.16	0.20	4.9	6.1
HMVC 12E	2 360	2 3239	18	2.4	4.5	4.9	1.5	0.20	0.20	5.1	6.2
HMVC 13E	2 548	2 5119	18	2.6	4.8	5.1	1.5	0.20	0.20	5.6	6.6
HMVC 14E	2 751	2 7149	18	2.8	5.0	5.3	1.5	0.20	0.20	5.9	7.1
HMVC 15E	2 933	2 8789	12	3.0	5.2	5.5	1.5	0.20	0.20	6.2	7.5
HMVC 16E	3 137	3 0829	12	3.2	5.4	5.7	1.5	0.20	0.20	6.5	8.2
HMVC 17E	3 340	3 2859	12	3.4	5.6	5.9	1.5	0.20	0.20	6.8	8.3
HMVC 18E	3 527	3 4729	12	3.6	5.8	6.1	1.5	0.20	0.20	7.3	8.8
HMVC 19E	3 730	3 6759	12	3.8	6.0	6.4	1.5	0.20	0.20	7.6	9.5
HMVC 20E	3 918	3 8639	12	4.0	6.2	6.5	1.5	0.24	0.20	7.9	9.7
HMVC 21E	4 122	4 0679	12	4.2	6.4	6.8	1.5	0.24	0.20	8.2	10.3
HMVC 22E	4 325	4 2709	12	4.4	6.7	7.0	1.5	0.24	0.20	8.7	10.9
HMVC 24E	4 716	4 6619	12	4.7	7.0	7.4	1.5	0.24	0.20	9.3	11.6
HMVC 26E	5.106	5 0519	12	5.1	7.5	7.8	1.5	0.24	0.20	9.9	12.5
HMVC 28E	5 497	5 4429	12	5.5	7.9	8.2	1.5	0.28	0.20	10.5	13.2
HMVC 30E	5 888	5 8339	12	5.9	8.3	8.7	1.5	0.28	0.20	11.6	14.6
HMVC 32E	6 284	6 2028	8	6.3	8.8	9.1	1.6	0.28	0.24	13.3	16.8
HMVC 34E	6 659	6 5778	8	6.7	9.3	9.6	1.6	0.28	0.24	14.6	18.5
HMVC 36E	7 066	6 9848	8	7.1	9.7	10.1	1.6	0.28	0.24	16.0	20.2
HMVC 38E	7 472	7 3908	8	7.5	10.2	10.6	1.7	0.31	0.28	17.8	23.1
HMVC 40E	7 847	7 7658	8	7.9	10.7	11.1	1.7	0.31	0.31	19.4	25.4
HMVC 44E	8 628	8 5468	8	8.7	11.5	12.0	1.7	0.31	0.35	22.3	29.8
HMVC 46E	9 125	9 0440	8	9.1	12.0	12.5	1.8	0.31	0.35	24.0	31.9
HMVC 48E	9 442	9 3337	6	9.5	12.4	13.0	1.8	0.35	0.39	25.6	35.3
HMVC 52E	10 192	10 0837	6	10.3	13.4	14.0	1.9	0.35	0.43	29.1	41.9
HMVC 56E	11 004	10 8957	6	11.1	14.3	15.0	1.9	0.35	0.47	32.7	48.5
HMVC 60E	11 785	11 6767	6	11.9	15.2	15.9	2.0	0.39	0.55	36.6	56.2
HMVC 64E	12 562	12 4537	6	12.7	16.1	16.9	2.1	0.39	0.55	40.8	65.0
HMVC 68E	13 339	13 2190	5	13.5	16.9	17.7	2.1	0.39	0.55	44.0	69.4
HMVC 72E	14 170	14 0500	5	14.3	17.9	18.6	2.2	0.39	0.59	48.5	78.3
HMVC 76E	14 957	14 8370	5	15.0	18.7	19.6	2.3	0.43	0.63	51.9	89.3
HMVC 80E	15 745	15 6250	5	15.8	19.6	20.6	2.4	0.43	0.67	56.9	100
HMVC 84E	16 532	16 4120	5	16.6	20.6	21.5	2.4	0.43	0.67	62.0	110
HMVC 88E	17 319	17 1990	5	17.4	21.4	22.3	2.4	0.47	0.67	65.9	119
HMVC 92E	18 107	17 9870	5	18.2	22.2	23.3	2.5	0.47	0.67	69.9	132
HMVC 96E	18 894	18 7740	5	19.0	23.1	24.1	2.6	0.47	0.75	75.3	139
HMVC 100E	19 682	19 5620	5	19.8	24.0	25.0	2.6	0.47	0.75	79.8	154
HMVC 106E	20 867	20 7220	4	20.9	25.4	26.4	2.7	0.51	0.83	87.1	174
HMVC 112E	22 048	21 9030	4	22.1	26.7	27.7	2.8	0.51	0.87	94.9	194
HMVC 120E	23 623	23 4780	4	23.7	28.4	29.4	2.9	0.51	0.91	104.3	220
HMVC 126E	24 804	24 6590	4	24.9	29.7	30.8	2.9	0.55	0.91	113.0	243
HMVC 134E	26 379	26 2340	4	26.5	31.3	32.5	3.0	0.55	0.94	123.2	265
HMVC 142E	27 961	27 7740	3	28.0	33.1	34.3	3.1	0.59	0.98	135.9	298
HMVC 150E	29 536	29 3490	3	29.6	34.8	35.9	3.1	0.59	0.98	147.6	322
HMVC 160E	31 504	31 3170	3	31.6	36.9	38.0	3.1	0.63	0.98	161.0	355
HMVC 170E	33 473	33 2860	3	33.5	39.0	40.2	3.3	0.63	1.02	177.6	399
HMVC 180E	35 441	35 2540	3	35.5	41.1	42.3	3.4	0.67	1.18	192.4	452
HMVC 190E	37 410	37 2230	3	37.5	43.2	44.3	3.4	0.67	1.18	210.3	481



TMEM 1500 (page 27)

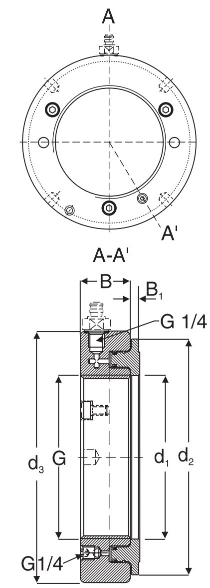
Designation	TMEM 1500		
Range of measurement	0 to 1,500 o/oo	**Operating temperature range**	–10 °C to 50 °C (14 °F to 122 °F)
Power supply	9–volt alkaline battery, type IEC 6LR61	**Accuracy**	+/- 1%, +/- 2 digits
Battery life	8 hours, continuous use	**IP rating**	IP 40
Low battery warning	Display shows "batt"	**Weight**	250 g (8.75 oz.)
Auto shut–off	After 30 minutes of inactivity	**Size**	157 × 84 × 30 mm (6.1 × 3.3 × 1.8 in)
Display	4–digit LCD with fixed decimal		

HMV E/A101 series (page 26 and 109)

Designation	HMV E/A101		
Mounting fluid	LHMF 300	**HMV 54E/A101 – HMV 92E/A101**	TMJL 100 / 728619 E/ TMJL 50
Recommended Pumps		**HMV 94E/A101 – HMV 200E/A101**	728619 E/ TMJL 50
HMV 10E/A101 – HMV 52E/A101	729124 / TMJL 100 / 728619 E/ TMJL 50	**Quick connection nipple**	729832 A (included)

Ordering details and dimensions

Designation	Bore diameter G mm	in	Designation	Bore diameter G mm	in	Designation	Bore diameter G mm	in
HMV 10E/A101	46,7	1.84	HMV 43E/A101	210,2	8.28	HMV 94E/A101	464,7	18.30
HMV 11E/A101	51,1	2.01	HMV 44E/A101	215,2	8.47	HMV 96E/A101	474,7	18.69
HMV 12E/A101	56,1	2.21	HMV 45E/A101	220,2	8.67	HMV 98E/A101	484,7	19.08
HMV 13E/A101	61,1	2.41	HMV 46E/A101	225,2	8.87	HMV 100E/A101	494,7	19.48
HMV 14E/A101	66,1	2.60	HMV 47E/A101	230,2	9.06	HMV 102E/A101	503,7	19.83
HMV 15E/A101	71,1	2.80	HMV 48E/A101	235,2	9.26	HMV 104E/A101	513,7	20.22
HMV 16E/A101	76,1	3.00	HMV 50E/A101	245,2	9.65	HMV 106E/A101	523,7	20.62
HMV 17E/A101	81,1	3.19	HMV 52E/A101	255,2	10.05	HMV 108E/A101	533,7	21.01
HMV 18E/A101	86,1	3.39	HMV 54E/A101	265,2	10.44	HMV 110E/A101	543,7	21.41
HMV 19E/A101	91,1	3.59	HMV 56E/A101	275,2	10.83	HMV 112E/A101	553,7	21.80
HMV 20E/A101	96,1	3.78	HMV 58E/A101	285,2	11.23	HMV 114E/A101	563,7	22.19
HMV 21E/A101	101,1	3.98	HMV 60E/A101	295,2	11.62	HMV 116E/A101	573,7	22.59
HMV 22E/A101	106,1	4.18	HMV 62E/A101	304,7	12.00	HMV 120E/A101	593,7	23.37
HMV 23E/A101	111,1	4.37	HMV 64E/A101	314,7	12.39	HMV 126E/A101	623,7	24.56
HMV 24E/A101	116,1	4.57	HMV 66E/A101	324,7	12.78	HMV 130E/A101	643,7	25.34
HMV 25E/A101	121,1	4.77	HMV 68E/A101	334,7	13.18	HMV 134E/A101	663,7	26.13
HMV 26E/A101	126,1	4.96	HMV 69E/A101	339,7	13.37	HMV 138E/A101	683,7	26.92
HMV 27E/A101	131,1	5.16	HMV 70E/A101	344,7	13.57	HMV 142E/A101	702,7	27.67
HMV 28E/A101	136,1	5.36	HMV 72E/A101	354,7	13.96	HMV 150E/A101	742,7	29.24
HMV 29E/A101	141,1	5.56	HMV 73E/A101	359,7	14.16	HMV 160E/A101	792,7	31.21
HMV 30E/A101	146,1	5.75	HMV 74E/A101	364,7	14.36	HMV 170E/A101	842,7	33.18
HMV 31E/A101	149,8	5.90	HMV 76E/A101	374,7	14.75	HMV 180E/A101	892,7	35.15
HMV 32E/A101	154,8	6.09	HMV 77E/A101	379,7	14.95	HMV 190E/A101	941,7	37.07
HMV 33E/A101	159,8	6.29	HMV 80E/A101	394,7	15.54	HMV 200E/A101	991,7	39.04
HMV 34E/A101	164,8	6.49	HMV 82E/A101	404,7	15.93			
HMV 36E/A101	174,8	6.88	HMV 84E/A101	414,7	16.33			
HMV 38E/A101	184,8	7.28	HMV 86E/A101	424,7	16.72			
HMV 40E/A101	194,8	7.67	HMV 88E/A101	434,7	17.11			
HMV 41E/A101	200,2	7.88	HMV 90E/A101	444,7	17.51			
HMV 42E/A101	205,2	8.08	HMV 92E/A101	454,7	17.90			



Feeler gauges 729865 series (page 27)

Designation	Blade length mm	in	Blade thickness mm	in	mm	in	mm	in
729865 A	100	4.0	0,03	0.0012	0,08	0.0031	0,14	0.0055
			0,04	0.0016	0,09	0.0035	0,15	0.0059
			0,05	0.0020	0,10	0.0039	0,20	0.0079
			0,06	0.0024	0,12	0.0047	0,30	0.0118
			0,07	0.0028				
729865 B	200	8.0	0,05	0.0020	0,18	0.0071	0,60	0.0236
			0,09	0.0035	0,19	0.0075	0,65	0.0256
			0,10	0.0039	0,20	0.0079	0,70	0.0276
			0,11	0.0043	0,25	0.0098	0,75	0.0295
			0,12	0.0047	0,30	0.0118	0,80	0.0315
			0,13	0.0051	0,35	0.0138	0,85	0.0335
			0,14	0.0055	0,40	0.0157	0,90	0.0354
			0,15	0.0059	0,45	0.0177	0,95	0.0374
			0,16	0.0063	0,50	0.0197	1,00	0.0394
			0,17	0.0067	0,55	0.0216		

TMJL 100 (page 29)

Designation	TMJL 100
Maximum pressure	100 MPa (14 500 psi)
Volume/stroke	1,0 cm^3 (0.06 in^3)
Oil container capacity	800 cm^3 (48 in^3)
Length of pressure hose	3 000 mm (118 in)
Connection nipple	G 1/4 quick connection
Weight	13 kg (29 lb)

TMJL 50 (page 30)

Designation	TMJL 50
Maximum pressure	50 MPa (7 250 psi)
Volume/stroke	3,5 cm^3 (0.21 in^3)
Oil container capacity	2 700 cm^3 (165 in^3)
Length of pressure hose	3 000 mm (118 in)
Connection nipple	G 1/4 quick connection
Weight	12 kg (26 lb)

Technical data

728619 E (page 30)

Designation	728619 E		
Maximum pressure	150 MPa (21 750 psi)	Oil container capacity	2 550 cm³ (155 in³)
Volume/stroke 1st Stage	20 cm³ below 2,5 MPa (1.2 in³ below 362 psi)	Length of pressure hose	3 000 mm (118 in)
Volume/stroke 2nd Stage	1 cm³ above 2,5 MPa (0.06 in³ above 362 psi)	Connection nipple	G 1/4 quick connection
			11,4 kg (25 lb)

THAP series (page 31)

Designation	Nominal hydraulic pressure	Pressure ratio	Maximum air pressure	Volume/ stroke	Oil outlet	Length	Height	Width	Weight
THAP 030	30 MPa 4 350 psi	1 : 59	0,7 MPa 101.5 psi	6,63 cm³ 0.40 in³	G 3/4	380 mm 15 in	190 mm 7.5 in	120 mm 4.7 in	21 kg 46.2 lb
THAP 030/SET	Complete set consisting of pump, high pressure hose and connecting nipples.								23 kg 50.7 lb
THAP 150	150 MPa 21 750 psi	1 : 252	0,7 MPa 101.5 psi	1,09 cm³ 0.06 in³	G 3/4	330 mm 13.0 in	190 mm 7.5 in	120 mm 4.7 in	19 kg 41.8 lb
THAP 150/SET	Complete set consisting of pump, pressure gauge, adapter block, high pressure hose and connecting nipples.								24 kg 52.9 lb
THAP 300E	300 MPa 43 500 psi	1 : 500	0,7 MPa 101.5 psi	0,84 cm³ 0.05 in³	G 3/4	405 mm 16 in	202 mm 8 in	171 mm 6.7 in	24,5 kg 54 lb
THAP 300E/SET	Complete set consisting of pump, pressure gauge, high pressure pipe.								24,5 kg 54 lb
THAP 400E	400 MPa 58 000 psi	1 :600	0,7 MPa 101.5 psi	0,65 cm³ 0.039 in³	G 3/4	405 mm 16 in	202 mm 8 in	171 mm 6.7 in	13 kg 28.6 lb
THAP 400E/SET	Complete set consisting of pump, pressure gauge, high pressure pipe								24,5 kg 54 lb

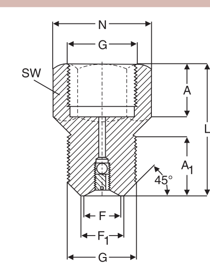
226270 and 226271 (page 31)

Injector	226270	226271
Valve nipple (optional)	226272	226273
Suitable shaft diameters	100 mm (4 in)	200 mm (8 in)
Maximum pressure	300 MPa (43 500 psi)	300 MPa (43 500 psi)
Oil container capacity	5,5 cm³ (0.33 in³)	25 cm³ (1.5 in³)
Connecting threads	G 3/8	G 3/4
Load to reach max. pressure	10 kg (22 lb)	30 kg (66 lb)
Weight	0,8 kg (1.8 lb)	2,1 kg (4.6 lb)

Valve nipple

Designation	Dimensions G	A mm	A in	A_1 mm	A_1 in	F mm	F in	F_1 mm	F_1 in	L mm	L in	N mm	N in
226272	G 3/8	15	0.59	17	0.67	9	0.35	10	0.39	40	1.57	25,4	1.00
226273	G 3/4	20	0.79	22	0.87	14	0.55	15	0.59	50	1.97	36,9	1.45

Designation	Width across flats mm	Width across flats in	Weight kg	Weight lb
226272	22	0.87	0,05	0.11
226273	32	1.26	0,20	0.44



TMJE series (page 33)

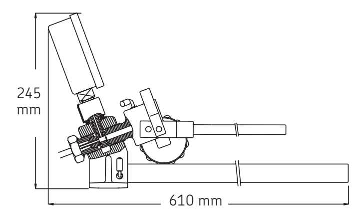
Injector set	TMJE 300	TMJE 400
Maximum pressure	300 MPa (43 500 psi)	400 MPa (58 000 psi)
Handle force at max. pressure	300 N (67.5 lbf)	400 N (90 lbf)
Volume per stroke	0,23 cm³ (0.014 in³)	0,23 cm³ (0.014 in³)
Oil container capacity	200 cm³ (12.2 in³)	200 cm³ (12.2 in³)
Weight	8 kg (18 lb)	8 kg (18 lb)
Pressure gauge	1077589	1077589/2
High pressure pipe	227957 A	227957 A/400MPa

226400 series (page 32)

Designation	226400	226400/400MPa
Maximum pressure	300 MPa (43,500 psi)	400 MPa (58,000 psi)
Volume /stroke	0,23 cm³ (0.014 in³)	0,23 cm³ (0.014 in³)
Oil container capacity	200 cm³ (12.2 in³)	200 cm³ (12.2 in³)
Connecting threads	G 3/4	G 3/4
Weight	2,2 kg (5 lb)	2,2 kg (5 lb)

226402 (page 33)

Designation	226402
Maximum pressure	400 MPa (58,000 psi)
Pressure gauge connection	G 1/2
Pressure pipe connection	G 3/4
Length of floor support	570 mm (22.4 in)
Weight	2,65 kg (6 lb)



High presure pipes (page 34)

Maximum working pressure	300 MPa (43 500 psi)	Outer pipe diameter	4 mm (0.16 in)
Test pressure	400 MPa (58 000 psi)	Inner pipe diameter	2 mm (0.08 in)
Test quantity	100%	Pipe lengths	Between 300 mm (12 in) and 4 000 mm (118 in) can be ordered e.g. 227957A/3000 (3000 mm long)

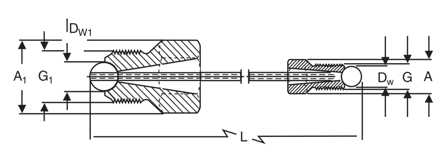
Ordering details and dimensions

Designation	G_1	G	A mm	A in	A_1 mm	A_1 in	Dw mm	Dw in	Dw_1 mm	Dw_1 in	L mm	L in	Weight kg	Weight lb
	–	–												
721740 A	G 3/4	G 1/8	11,5	0.45	36,9	1.45	7,94	0.31	15,88	0.63	1 000	39	0,3	0.7
227957 A*	G 3/4	G 1/4	17,3	0.68	36,9	1.45	11,11	0.44	15,88	0.63	2 000	78	0,4	0.9
227958 A*	G 3/4	G 3/4	36,9	1.45	36,9	1.45	15,88	0.63	15,88	0.63	2 000	78	0,6	1.3
1020612 A**	G 1/4	G 1/4	17,3	0.68	17,3	0.68	11,11	0.44	11,11	0.44	1 000	39	0,5	1.1
728017 A	G 1/4	G 1/4	17,3	0.68	17,3	0.68	11,11	0.44	7,94	0.31	300	12	0,2	0.4
727213 A***	G 1/4	G 1/4	17,3	0.68	17,3	0.68	7,94	0.31	7,94	0.31	300	12	0,2	0.4
729123 A	G 3/4	G 1/4	17,3	0.68	36,9	1.45	7,94	0.31	15,88	0.63	300	12	0,3	0.7

* *These pipes are also available in a 400 MPa execution.*
Designations are 227957 A/400MP and 227958 A/400MP.
Outer diameter of the pipe is 6 mm (0.24 in)

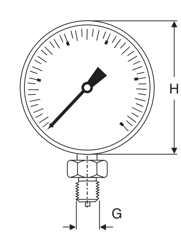
** *Maximum pressure 400 MPa (58 000 psi).*
Outer diameter of the pipe 6 mm (0.24 in).

*** *The high pressure pipe 727213 A is designed to fit small OK–couplings.*
This pipe is not suitable for normal oil injection connection holes.



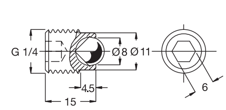
Pressure gauges (page 34)

Designation	Pressure range MPa	Pressure range psi	Diameter H mm	Diameter H in	Connection thread	Weight kg	Weight lb	Accuracy % of full scale
1077587	0 – 100	0 – 14 500	100	3.94	G 1/2	0,80	1.8	1
1077587/2	0 – 100	0 – 14 500	63	2.48	G 1/4	0,25	0.6	1,6
TMJG 100D	0 – 100	0 – 15 000	76	3.00	G 1/4	0,21	0.5	<0,2
1077589	0 – 300	0 – 43 500	100	3.94	G 1/2	0,80	1.8	1
1077589/2	0 – 400	0 – 58 000	100	3.94	G 1/2	0,80	1.8	1

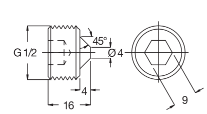


Plugs for oil ducts and vent holes (page 34)

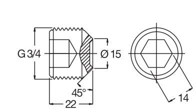
Designation	Thread	Length mm	Length in	Weight kg	Weight lb	Suitable hex. key mm	Suitable hex. key in
233950 E	G 1/4	15	0.59	0,02	0.04	6	0.24
729944 E	G 1/2	17	0.67	0,03	0.07	9	0.35
1030816 E	G 3/4	23	0.90	0,05	0.11	14	0.55



Plug 233950 E



Plug 729944 E



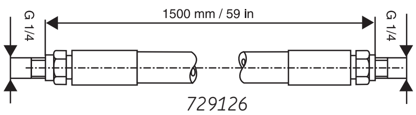
Plug 1030816 E

Maximum working pressure 400 MPa (58 000 psi)

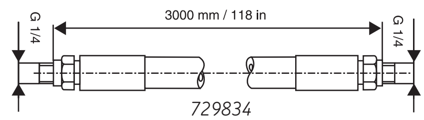
Technical data

Flexible high pressure hoses (page 34)

Designation	Bore diameter		Outside diameter		Maximum working pressure		Minimum burst pressure		Minimum bending radius		End fittings	Working temperature		Length		Weight	
	mm	in	mm	in	MPa	psi	MPa	psi	mm	in		°C	°F	mm	in	kg	l
729126	4,0	0.16	10	0.39	100	14 500	300	43 500	65	2.6	G 1/4	–30/80	–22/176	1 500	59	0,4	0.9
729834	5,0	0.20	11	0.43	150	21 750	450	65 250	150	5.9	G 1/4	–30/80	–22/176	3 000	118	0,9	2.0



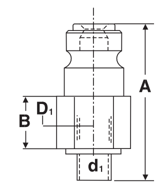
729126



729834

Quick connection coupling and nipples (page 35)

Designation	Thread d_2	Dimensions D_2		C		A		Maximum pressure	
Coupling		mm	in	mm	in	mm	in	MPa	psi
729831 A	G 1/4	24	0.94	27	1,06	58	2.28	150	21 750
Nipples	d_1	D_1		B		A			
729832 A	G 1/4	22	0.87	14	0.55	46	1.81	150	21 750
729100	G 1/8	17	0.67	14	0.55	43	1.69	100	14 500



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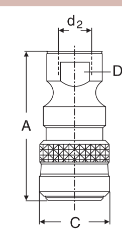
729100

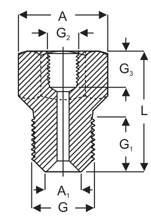


729831A

Connection nipples with NPT tapered threads (page 35)

Designation	G	G_2	Dimensions A		G_1		G_3		L		Key width		Weight	
			mm	in	mm	in	mm	in	mm	in	mm	in	kg	lb
729654	NPT 1/4"	G 1/4	25,4	1.00	15	0.59	15	0.59	42	1.65	22	0.87	0,25	0.55
729655	NPT 3/8"	G 1/4	25,4	1.00	15	0.59	15	0.59	40	1.57	22	0.87	0,25	0.55
729106	G 1/4	NPT 3/8"	36,9	1.45	17	0.67	15	0.59	50	1.97	32	1.26	0,16	0.35
729656	NPT 3/4"	G 1/4	36,9	1.45	20	0.79	15	0.59	45	1.77	32	1.26	0,30	0.66

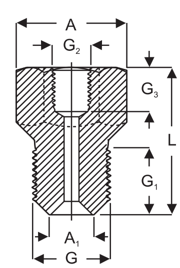
Maximum working pressure 300 MPa (43,500 psi)



Connection nipples with metric and G pipe threads (page 35)

Designation	G	G_2	Dimensions A		A_1		G_1		G_3		L	
			mm	in	mm	in	mm	in	mm	in	mm	in
1077456	M 8	M 6	11	0.43	5	0.20	15	0.59	9	0.35	33	1.30
1077455	G 1/8	M 6	11	0.43	7	0.28	15	0.59	9	0.35	33	1.30
1014357 A	G 1/8	G 1/4	25,4	1.00	7	0.28	15	0.59	15	0.59	43	1.69
1009030 B	G 1/8	G 3/8	25,4	1.00	7	0.28	15	0.59	15	0.59	42	1.65
1019950	G 1/8	G 1/2	36,9	1.45	7	0.28	15	0.59	14	0.55	50	1.97
1018219 E	G 1/4	G 3/8	25,4	1.00	9,5	0.37	17	0.67	15	0.59	45	1.77
1009030 E	G 1/4	G 3/4	36,9	1.45	9,5	0.37	17	0.67	20	0.79	54	2.13
1012783 E	G 3/8	G 1/4	25,4	1.00	10	0.39	17	0.67	15	0.59	43	1.96
1008593 E	G 3/8	G 3/4	36,9	1.45	10	0.39	17	0.67	20	0.79	53	2.09
1016402 E	G 1/2	G 1/4	25,4	1.00	14	0.55	20	0.79	15	0.59	43	1.96
729146	G 1/2	G 3/4	36,9	1.45	–	–	17	0.67	20	0.79	50	1.97
228027 E	G 3/4	G 1/4	36,9	1.45	15	0.59	22	0.87	15	0.59	50	1.97

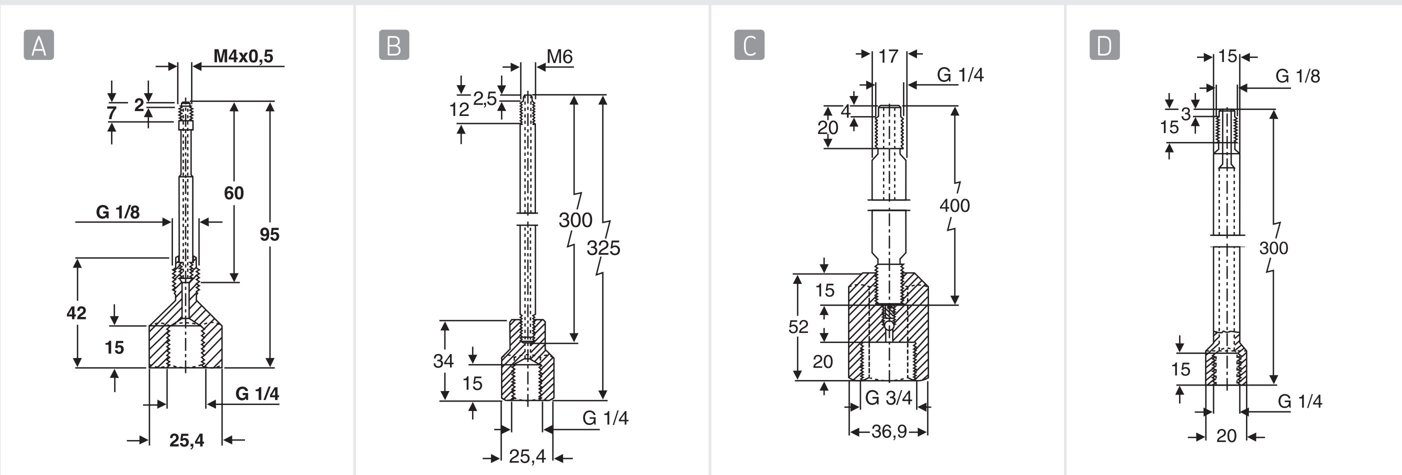
Designation	Width across flats		Weight	
	mm	in	kg	lb
1077456	10	0.39	0,05	0.11
1077455	10	0.39	0,05	0.11
1014357 A	22	0.87	0,06	0.13
1009030 B	22	0.87	0,06	0.13
1019950	32	1.26	0,14	0.31
1018219 E	22	0.87	0,07	0.15
1009030 E	32	1.26	0,13	0.29
1012783 E	22	0.87	0,08	0.18
1008593 E	32	1.26	0,15	0.33
1016402 E	22	0.87	0,10	0.22
729146	32	1.26	0,18	0.40
228027 E	32	1.26	0,25	0.55



All nipples with E suffix have a maximum working pressure of 400 MPa (58 000 psi), otherwise maximum working pressure is 300 MPa (43 500 psi)

Extension pipes with connection nipples (page 36)

M4 extension pipe with connection nipple (A)		
Designation	**pipe 234064**	**nipple 234063**
Max. pressure	50 MPa (7 250 psi)	50 MPa (7 250 psi)
M6 extension pipe with connection nipple (B)		
Designation	**pipe 1077453**	**nipple 1077454**
Max. pressure	200 MPa (29 000 psi)	200 MPa (29 000 psi)
Valve nipple with extension pipe (C)		
Designation	**pipe 227964**	**nipple 227963**
Max. pressure	300 MPa (43 500 psi)	300 MPa (43 500 psi)
Extension pipe (D)		
Designation	**227965**	
Max. pressure	300 MPa (43 500 psi) Normally used in combination with high pressure pipe eg. 227957 A	



TMBA G11W (page 38)

Designation	**TMBA G11W**
Size	9
Colour	White/blue
Pack size	1 pair

TMBA G11 (page 38)

Designation	**TMBA G11**
Material	Hytex
Inner lining	Cotton
Size	9
Colour	White
Maximum temperature	150 °C (302 °F)
Pack size	1 pair

TMBA G11ET (page 39)

Designation	**TMBA G11ET**
Material	KEVLAR®
Inner lining	Cotton
Size	10 (EN 420 size)
Colour	Yellow
Maximum temperature	500 °C (932 °F)
Pack size	1 pair

TMBA G11H (page 39)

Designation	**TMBA G11H**
Material	Polyaramid
Inner lining	Nitrile
Size	10
Colour	Blue
Maximum temperature	250 °C (482 °F)
Pack size	1 pair

TMEA P1 (page 46) (optional for TMEA 1P/2.5 and standard for TMEA 1PEx)

Printing system	Thermal dot matrix
Power	Rechargeable battery – 12V maximum, Continental European adapter
Operation time	60 minutes continuous operation with fully charged battery

Product and accessories ordering details

Designation	Description
TMEA 2	Shaft alignment tool
TMEA 1P/2.5	Shaft alignment tool with printer capability
TMEA 1PEx	Intrinsically safe shaft alignment with printer
TMEA P1	Thermal printer complete with Continental European adaptor and connection cable (TMEA 1P/2.5 and TMEA 1PEx only)
TMEA C2	Extension chain set (1 020 mm / 40.1 in)
TMEA F2	1 × non magnetic fixture, chain and 220 mm (8.6 in) rod
TMEA F6	2 thin chain fixtures, complete set
TMEA F7	Set of 3 pairs of connection rods; short: 150 mm (5.9 in), medium: 220 mm (8.6 in) and long: 320 mm (12.5 in)
TMEA MF	1 magnetic fixture
TMEA P1-10	UK / Australian mains adaptor for the printer
TMEA R1	3 spare rolls of thermal paper for the printer

Technical data

TMEA series (page 44 – 45)

Designation	TMEA 2	TMEA 1/P2.5	TMEA 1PEx
Measuring units:			
Type of laser	Diode laser	Diode laser	Diode laser
Laser wave length	670 – 675 nm	670 – 675 nm	670 – 675 nm
Laser class	2	2	2
Maximum laser power	1 mW	1 mW	1 mW
Maximum distance between measuring units	0,850 m (2,8 ft)	2,50 m (8,2 ft)	1 m (3 ft)
Type of detectors	Single axis PSD, 8,5 × 0,9 mm (0.3 × 0.4 in)	Single axis PSD, 10 × 10 mm (0.4 × 0.4 in)	Single axis PSD, 10 × 10 mm (0.4 × 0.4 in)
Fixture	Magnetic and/or chain	Chain standard Magnetic optional	Chain standard Magnetic optional
Display unit:			
Battery type	2 × 1,5V LR14 Alkaline	3 × 1,5V LR14 Alkaline	Special type of LR 14 batteries
Operating time	20 hours continuous operation	20 hours continuous operation	20 hours continuous operation
Displayed resolution	0,01 mm (0.1 mil in “inch” setting)	0,01 mm (0.1 mil in “inch” setting)	0,01 mm (0.1 mil in “inch” setting)
Complete system:			
Content	Display unit 2 measuring units with spirit levels 2 magnetic / mechanical shaft fixtures 2 locking chains 5 sets of shims Measuring tape Instructions for use Set of alignment reports Carrying case	Display unit 2 measuring units with spirit levels 2 mechanical shaft fixtures 2 locking chains 2 extension chains 5 sets of shims Measuring tape Instructions for use Set of alignment reports Carrying case	Display unit 2 measuring units with spirit levels 2 mechanical shaft fixtures 2 locking chains 2 extension chains 5 sets of shims Measuring tape Instructions for use Set of alignment reports Carrying case Printer Charger Connection cable Spare paper roll
Shaft diameter range	Magnetic: 40 – 500 mm (1.6 – 20 in) Chain: 40 – 150 mm (1.6 – 5,9 in) Optional chain: 150 – 500 mm (5.9 – 20 in)	30 – 500 mm (1.2 – 20 in)	30 – 500 mm (1.2 – 20 in)
Accuracy of system	Better than 2%	Better than 2%	Better than 2%
Ex classification	–	–	II 2 G, EEx ib IIC T4
Ex certificate number	–	–	Nemko03ATEX101X
Temperature range	0 – 40 °C (32 – 104 °F)	0 – 40 °C (32 – 104 °F) without printer	0 – 40 °C (32 – 104 °F) without printer
Operating humidity	< 90 %	< 90 % without printer	< 90 % without printer
Carrying case dimensions	390 × 340 × 95 mm (15.4 × 13.4 × 3.7 in)	534 × 427 × 157 mm (21.0 × 16.8 × 6.2 in)	534 × 427 × 157 mm (21.0 × 16.8 × 6.2 in)
Total weight (incl. case)	3,7 kg (8.1 lb)	8,9 kg (19.6 lb)	8,9 kg (19.6 lb)
Calibration certificate	Valid for two years	Valid for two years	Valid for two years
Warranty	12 months	12 months	12 months
Printing capability	No	Yes – printer is optional	Yes – printer is standard

TMEB 2 (page 48)

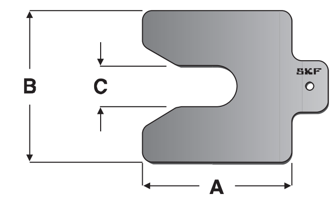
Designation	TMEB 2		
Content	1 laser unit 1 receiver unit 4 set of V guides Carrying case	Dimensions laser unit Dimensions receiver unit Battery type	70 × 74 × 61 mm (2.8 × 2.9 × 2.4 in) 96 × 74 × 61 mm (3.8 × 2.9 × 2.4 in) 2 × 1,5V LR03 (AAA) batteries in laser unit
Housing material	Extruded aluminium	Battery lifetime	20 hours continuous operation
Type of laser	Diode laser, class 2, 1 mW	Weight laser unit	320 g (11.3 oz)
Laser wave length	632 nm	Weight receiver unit	270 g (9.5 oz)
Measurement distance	50 mm to 6,000 mm (2 in to 20 ft)	Calibration certificate	Valid for two years
Fixture	Magnetic	Warranty	12 months
Measurement accuracy angular	Better than 0,2°		
Measurement accuracy linear	Better than 0,5 mm		

Accessory ordering details

Designation	Description
TMEB A2	2 × magnetic side adaptor for chain sprocket, multi–ribbed and timing belt pulleys
TMEB G2	Set of V–guides, 4 different sizes

TMAS series (page 46-47)

Designation	Number of shims per set	A mm	B mm	C mm	Thickness mm	Designation	Number of shims per set	A mm	B mm	C mm	Thickness mm
TMAS 50-005	10	50	50	13	0,05	TMAS 75-005	10	75	75	21	0,05
TMAS 50-010	10	50	50	13	0,10	TMAS 75-010	10	75	75	21	0,10
TMAS 50-020	10	50	50	13	0,20	TMAS 75-020	10	75	75	21	0,20
TMAS 50-025	10	50	50	13	0,25	TMAS 75-025	10	75	75	21	0,25
TMAS 50-040	10	50	50	13	0,40	TMAS 75-040	10	75	75	21	0,40
TMAS 50-050	10	50	50	13	0,50	TMAS 75-050	10	75	75	21	0,50
TMAS 50-070	10	50	50	13	0,70	TMAS 75-070	10	75	75	21	0,70
TMAS 50-100	10	50	50	13	1,00	TMAS 75-100	10	75	75	21	1,00
TMAS 50-200	10	50	50	13	2,00	TMAS 75-200	10	75	75	21	2,00
TMAS 50-300	10	50	50	13	3,00	TMAS 75-300	10	75	75	21	3,00
TMAS 100-005	10	100	100	32	0,05	TMAS 125-005	10	125	125	45	0,05
TMAS 100-010	10	100	100	32	0,10	TMAS 125-010	10	125	125	45	0,10
TMAS 100-020	10	100	100	32	0,20	TMAS 125-020	10	125	125	45	0,20
TMAS 100-025	10	100	100	32	0,25	TMAS 125-025	10	125	125	45	0,25
TMAS 100-040	10	100	100	32	0,40	TMAS 125-040	10	125	125	45	0,40
TMAS 100-050	10	100	100	32	0,50	TMAS 125-050	10	125	125	45	0,50
TMAS 100-070	10	100	100	32	0,70	TMAS 125-070	10	125	125	45	0,70
TMAS 100-100	10	100	100	32	1,00	TMAS 125-100	10	125	125	45	1,00
TMAS 100-200	10	100	100	32	2,00	TMAS 125-200	10	125	125	45	2,00
TMAS 100-300	10	100	100	32	3,00	TMAS 125-300	10	125	125	45	3,00
TMAS 200-005	10	200	200	55	0,05						
TMAS 200-010	10	200	200	55	0,10						
TMAS 200-020	10	200	200	55	0,20						
TMAS 200-025	10	200	200	55	0,25						
TMAS 200-040	10	200	200	55	0,40						
TMAS 200-050	10	200	200	55	0,50						
TMAS 200-070	10	200	200	55	0,70						
TMAS 200-100	10	200	200	55	1,00						
TMAS 200-200	10	200	200	55	2,00						
TMAS 200-300	10	200	200	55	3,00						



Kits of single slot shims (metric)

Designation	Contents	Weight
TMAS 340	340 shims in 9 thicknesses and 2 sizes	17 kg (37.4 lb)
TMAS 360	360 shims in 6 thicknesses and 3 sizes	12 kg (26.4 lb)
TMAS 510	510 shims in 9 thicknesses and 3 sizes	14 kg (30.8 lb)
TMAS 720	720 shims in 9 thicknesses and 4 sizes	30 kg (66 lb)

LAGD 1000 (page 71)

Designation	LAGD 1000/B	LAGD 1000/DC	LAGD 1000/AC
Max. operating pressure	150 bars (2 175 psi)	150 bars (2 175 psi)	150 bars (2 175 psi)
Permissible operating temperature	–10 to 60 °C (14 to 140 °F)	–25 to 75 °C (–13 to 167 °F)	–25 to 60 °C (–13 to 140 °F)
Number of outlets	6 to 12	10 to 20	10 to 20
Max. length of pipes	6 m (19.7 ft)	6 m (19.7 ft)	6 m (19.7 ft)
Tubing	6 x 1,25 mm (0.05 in)	6 x 1,25 mm (0.05 in)	6 x 1,25 mm (0.05 in)
Output of pump element	1 cm^3/min (0.061 in^3/min)	2 cm^3/min (0.122 in^3/min)	2 cm^3/min (0.122 in^3/min)
Reservoir capacity	1 litre (33.8 US fl. oz)	1 litre (33.8 US fl. oz)	1 litre (33.8 US fl. oz)
Greases	up to NLGI grade 2 Flow pressure < 300 mbar	up to NLGI grade 2 Flow pressure < 700 mbar	up to NLGI grade 2 Flow pressure < 700 mbar
Weight	5,8 kg (12.8 lbs)	3,7 kg (8.2 lbs)	4,8 kg (10.6 lbs)
System of protection	IP65	IP65	IP65
Electrical specifications			
Power connection	n/a	DIN EN 175 301-803, plug supplied	DIN EN 175 301-803, plug supplied
Rated voltage	18V	24V DC	110 - 240V 50/60 Hz
Power consumption	16 Ah	n/a	n/a
Battery type	alkaline	n/a	n/a
Type power input at 20 °C (68 °F) and max. operating pressure		0,5 A	1,3A / 110V 0,4A / 230V
Battery pack life	12 months or 1 lubricator filling (which ever comes first), when installed by end of the battery pack expiry date.		

VKN 550 (page 80)

Designation	VKN 550		
Description	Bearing grease packer	Other greases	NLGI class 000 to 2
Weight	1,8 kg (3.9 lb)	Bearing range	
Material	Zinc plated, metal finish	– Inner diameter d	19 to 120 mm
Suitable greases	approved for all SKF greases	– Outer diameter D	Max 200 mm

Technical data

Bearing greases (page 60 – 67)	LGMT 2	LGMT 3	LGEP 2	LGFP 2	LGFB 2	LGEM 2
DIN 51825 code	K2K–30	K3K–30	KP2G–20	K2G–20	KPE/HC2K-20L	K2G–20
NLGI consistency class	2	3	2	2	2	2
Soap type	Lithium	Lithium	Lithium	Aluminium complex	Aluminium complex	Lithium
Colour	Red brown	Amber	Light brown	Transparent	Beige	Black
Base oil type	Mineral	Mineral	Mineral	Medical white oil	PAO/Ester	Mineral
Operating temperature range	–30 to 120 °C (–22 to 250 °F)	–30 to 120 °C (–22 to 250 °F)	–20 to 110 °C (–4 to 230 °F)	–20 to 110 °C (–4 to 230 °F)	–25 to 120 °C (–13 to 248 °F)	–20 to 120 °C (–4 to 250 °F)
Dropping point DIN ISO 2176	>180 °C (>356 °F)	>180 °C (>356 °F)	>180 °C (>356 °F)	>250 °C (>482 °F)	>220 °C (>428 °F)	>180 °C (>356 °F)
Base oil viscosity:						
40 °C, mm^2/s	110	120–130	200	130	266	500
100 °C, mm^2/s	11	12	16	7,3	29	32
Penetration DIN ISO 2137:						
60 strokes, 10^{-1} mm	265 – 295	220 – 250	265 – 295	265 – 295	265 – 295	265 – 295
100 000 strokes, 10^{-1} mm	+50 max. (325 max.)	280 max.	+50 max. (325 max.)	+30 max.	+50 max.	325 max.
Mechanical stability:						
Roll stability, 50 hrs at 80 °C, 10^{-1} mm	+ 50 max.	295 max.	+50 max.		29* (2 hrs at 70 °C)	345 max.
Roll stability, 72 hrs at 100 °C, 10^{-1} mm						
SKF V2F test	'M'	'M'	'M'			'M'
Corrosion protection:						
SKF Emcor: – standard ISO 11007	0 – 0	0 – 0	0 – 0	0 – 0	0 – 0	0 – 0
– water washout test	0 – 0	0 – 0	0 – 0			0 – 0
– salt water test (100% seawater)	0 – 1*		1 – 1*			
Water resistance DIN 51 807/1, 3 hrs at 90 °C	1 max.	2 max.	1 max.	1 max.	1 max.	1 max.
Oil separation DIN 51 817, 7 days at 40 °C, static, %	1 – 6	1 – 3	2 – 5	1 – 5	1 – 5 max.	1 – 5
Lubrication ability SKF R2F, running test B at 120 °C	Pass	Pass	Pass			Pass at 100 °C (212 °F)
Copper corrosion DIN 51 811, 110 °C	2 max. (130 °C / 266 °F)	2 max.	2 max. (100 °C)		2 max.	2 max.
Rolling bearing grease life SKF R0F test L50 life at 10 000 rpm, hrs		1 000 min. at 130 °C (266 °F)		1 000 at 110 °C (230 °F)	1 000 min. at 110 °C (230 °F)	
EP performance						
Wear scar DIN 51350/5, 1 400 N, mm			1,4 max		1,6 max	1,4 max.
4–ball test, welding load DIN 51350/4			2 800 min.	1 100 min.	1 800 N min.	3 000 min.
Fretting corrosion ASTM D4170 (mg)			5,7 *			
Available pack sizes	35, 200 g tube 420 ml cart. 1, 5, 18, 50, 180 kg –	– 420 ml cart. 0,5, 1, 5, 18, 50, 180 kg –	– 420 ml cart. 1, 5, 18, 50, 180 kg –	420 ml cart. 1, 18, 180 kg SYSTEM 24 (LAGD / LAGE)	420 ml cart. 1, 18, 180 kg	420 ml cart. 5, 18, 180 kg SYSTEM 24 (LAGD / LAGE)
Designation	LGMT 2/ (pack size)	LGMT 3/ (pack size)	LGEP 2/ (pack size)	LGFP 2/ (pack size)	LGFB 2/ (pack size)	LGEM 2/ (pack size)

** Typical value*

LGEV 2	LGLT 2	LGFL 1	LGGB 2	LGWM 1	LGWA 2	LGHB 2	LGHP 2	LGET 2
KPF2K–10	KP2G–50	KE/HC 1K-40	KPE 2K–40	KP1G–30	KP2N–30	KP2N–20	K2N–40	KFK2U–40
2	2	1	2	1	2	2	2 –3	2
Lithium / calcium	Lithium	Aluminium complex	Lithium / calcium	Lithium	Lithium complex	Complex calcium sulphonate	Di–urea	PTFE
Black	Beige	White	Off white	Brown	Amber	Brown	Blue	Whitish cream
Mineral	PAO	PAO/Ester	Synthetic ester	Mineral	Mineral	Mineral	Mineral	Synthetic (fluorinated polyether)
–10 to 120 °C (14 to 250 °F)	–50 to 110 °C (–58 to 230 °F)	–45 to 120 °C (–49 to 248 °F)	–40 to 120 °C (–40 to 250 °F)	–30 to 110 °C (–22 to 230 °F)	–30 to 140 °C (-22 to 284 °F)	–20 to 150 °C (–4 to 300 °F)	–40 to 150 °C (–40 to 300 °F)	–40 to 260 °C (–40 to 500 °F)
>180 °C (>356 °F)	>180 °C (>356 °F)	>220 °C (>428 °F)	> 170 °C (<338 °F)	>170 °C (338) °F	>250 °C (482 °F)	>220 °C (428 °F)	>240 °C (464 °F)	>300 °C (572 °F)
1 020 58	18 4,5	30 6	110 13	200 16	185 15	400 – 450 26,5	96 10,5	400 38
265 – 295 325 max.	265 – 295 +50 max.	310 – 340 +50 max.	265 – 295 +50 max. (325 max.)	310 – 340 +50 max.	265 – 295 +50 max. (325 max.)	265 – 295 -20 – +50 (325 max.)	245 – 275 (365 max.)	265 – 295
+70 max. +50 max. 'M'	+380 max.		+70 max. (350 max.)		+50 max. change 'M'	–20 – +50 change 'M'	365 max.	± 30 max. (130 °C/266 °F)
0 – 0 0 – 0* 0 – 0*	0 – 1	0 – 0	0 – 0	0 – 0 0 – 0	0 – 0 0 – 0*	0 – 0 0 – 0 0 – 0*	0 – 0 0 – 0 0 – 0	1 – 1
1 max.	1 max.	1 max.	0 max.	1 max.	1 max.	1 max.	1 max.	0 max.
1 – 5	<4	15 max.	0,3 – 3	8 – 13	1 – 5	1 – 3 (at 60 °C)	1 – 5	13 max. (30 hrs at 200 °C)
			Pass at 100 °C* (212 °F)		Pass at 100 °C (212 °F)	Pass at 140 °C (284 °F)	Pass	
1 max (90 °C/194 °F)	1 max. (150 °C/300 °F)	2 max.		2 max.	2 max. (150 °C/300 °F)	2 max. (150 °C/300 °F)	1 max.	1
	> 1 000, 20 000 rmp at 100 °C (212 °F)	1 000 min. at 110 °C (230 °F)	>300 at 120 °C (250 °F)			>1 000 at 130 °C (266 °F)	1 000 min. at 150 °C (302 °F)	>700, 5 600 rpm* at 220 °C (428 °F)
1,2 max. 3 000 min.	2 000 min.		1,8 max. 2 600 min.	1,8 max. 3 200 min.*	1,6 max. 2 600 min.	0,86*. 4 800 N*		8 000 min.
				5,5 *		0*	7*	
35 g tube 420 ml cart. 5, 18, 50, 180 kg	200 g tube 1, 25, 180 kg	1, 18, 180 kg	420 ml cart. 5, 18, 180 kg SYSTEM 24 (LAGD)	420 ml cart. 5, 50, 180 kg	35, 200 g tube 420 ml cart. 1, 5, 50, 180 kg SYSTEM 24 (LAGD / LAGE)	420 ml cart. 5, 18, 50, 180 kg SYSTEM 24 (LAGD / LAGE)	420 ml cart. 1, 5, 18, 50, 180 kg SYSTEM 24 (LAGD / LAGE)	50 g (25 ml) Syringe 1 kg
LGEV 2/ (pack size)	LGLT 2/ (pack size)	LGFL 1/ (pack size)	LGGB 2/ (pack size)	LGWM 1/ (pack size)	LGWA 2/ (pack size)	LGHB 2/ (pack size)	LGHP 2/ (pack size)	LGET 2/ (pack size)

Technical data

LAGD series (page 68 – 69)

Grease capacity	**LAGD 60**	**60 ml (2.03 fl oz. US)**
	LAGD 125	**125 ml (4.25 fl oz. US)**

Nominal emptying time	Adjustable; 1 – 12 months
Ambient temperature range	
LAGD 60/.. and LAGD 125/..	–20 to 60 °C (–5 to 140 °F)
LAGD 125/F..	–20 to 55 °C (–5 to 131 °F)
Maximum operating pressure	5 bar (75 psi) (at start-up)
Drive mechanism	Gas cell producing inert gas
Connection thread	R 1/4
Maximum feed line length with:	
– grease	300 mm (11.8 in)
– oil	1 500 mm (59.1 in)
Intrinsically safe approval	II 1 G Ex ia IIC T6 II 1 D Ex iaD 20 T85°C I M1 Ex ia I
EC Type Examination Certificate	
LAGD 60/.. and LAGD 125/..	Kema 04ATEX1275 X Issue 2
LAGD 125/F..	Kema 07ATEX0132 X
Protection class	IP 68
Recommended storage temperature	20 °C (70 °F)
Storage life of lubricator	2 years
Weight	LAGD 125 approx 200 g (7.1 oz) LAGD 60 approx 130 g (4.6 oz) Lubricant included

LAGE series (page 70 – 71)

Grease capacity	**LAGE 125**	**122 ml (4.1 fl. oz US)**
	LAGE 250	**250 ml (8.5 fl. oz US)**

Emptying time	User adjustable: 1, 3, 6, 9 and 12 months
Ambient temperature range	0 °C (-10°C peak) to 50 °C (32 °F (14 °F peak) to 122 °F)
Maximum operating pressure	5 bar (75 psi)
Drive mechanism	Electro mechanical
Connection thread	R 1/4
Maximum feed line length with:	
– grease	Up to 3 meters (10 ft) *
– oil	Up to 5 meters (16 ft)
LED status indicators	operating, purging lubricant, empty, malfunction
UL certification	UL listed T code 59°C - Category BAYZ – 92UM Lubricant dispensing equipment for use in hazardous locations Class I, Division II, Group A, B, C, D Class II, Division II, Group F &G Class III
Protection class assembled lubricator	IP 65
Battery pack	4,5V 2,7 Ah - Alkaline manganese
Recommended storage temperature	20 °C (70 °F)
Storage life of lubricator	3 years ** (2 years for LGFP 2 and Oils)
Total weight	
– LAGE 125	635 g (22.5 oz)
– LAGE 250	800 g (28.2 oz)

* *The maximum feed line length is dependent on ambient temperature, grease type and back pressure created by the application.*

** *Storage life is 3 years from production date, which is printed on the side of the canister. The canister and battery pack may be used even at 12 months setting if activated 3 years from production date.*

LAGD 400 (page 74)

Designation	**LAGD 400**
Content	8–outlet lubricator 20 m tubing Quick connectors for application side 2 Y–connectors LGMT 2/0.4 grease cartridge SKF's DialSet program
Number of feed lines	1 – 8
Maximum pressure	40 bar (600 psi)
Suitable grease	NLGI 1, 2 and 3
Maximum length of feed–lines	5 m (16 ft)
Ambient temperature	0 – 50 °C (32 – 120 °F)
Drive mechanism	Electro–mechanical
Volume	0,1 – 10 cm³/day (0.003 – 0.35 US fl. oz/day) per outlet approx 0,6 – 65 g/week (0,02 – 2,3 oz/week)
Power	110–240V AC, 50–60Hz or 24V DC
Alarms	Blocked feed lines, empty cartridge; internal and external
External steering	External relay steering
IP rating	54
Lubrication tubes	20 m (65 ft), Nylon, 6 × 1,5 mm (1/4 × 0.06 in)
Connection thread	G 1/4
Height	530 mm (21 in)

LAGM 1000E (page 80)

Designation	**LAGM 1000E**
Housing material	Aluminium, anodised
Weight	0,3 kg (0.66 lb)
IP rating	IP 67
Suitable greases	NLGI 0 – NLGI 3
Maximum operating pressure	70 MPa (10 000 psi)
Maximum grease flow	1 000 cm³/min (34 US fl. oz/min)
Thread connections	M10 × 1
Display	Lit LCD (4 digits / 9 mm)
Accuracy	±3% from 0 – 300 bar ±5% from 300 – 700 bar
Selectable units	cm³, g, US fl. oz or oz
Display lamp auto switch off	15 seconds after last pulse
Low battery	Indication on display
Battery type	1,5 V LR1 (2×) Alkaline
Unit auto switch off	1 minute after last pulse

Chemicals and oils LHRP 1 (page 38) LGAF 3E (page 10) LHMT 68, LHHT 265, LHFP 150 (page 73)

	LHRP 1	LGAF 3E	LHMT 68	LHHT 265	LHFP 150
Description	Anti–corrosive agent	Anti–fretting paste	Medium temperature oil	High temperature oil	Food compatible, NSF H1 oil
Specific gravity	0,815	1,19	0,85	0,91	0,85
Colour	Hazy brown	White–beige	Yellow–brown	Yellow–orange	Colourless
Base oil type	Mineral	Mineral and synthetic	Mineral	Synthetic ester	Synthetic ester
Thickener	Not applicable	Lithium soap	Not applicable	Not applicable	Not applicable
Operating temperature range, °C (°F)	– –	–25 to 250 °C (–13 to 482 °F)	–15 to 90 °C (5 to 194 °F)	Up to 250 °C (482 °F)	–30 to 120 °C (–22 to 248 °F)
Base oil viscosity: 20 °C, mm²/s 40 °C, mm²/s 100 °C, mm²/s	 not valid because of thixotropic nature	 – 17,5 –	 – ISO VG 68 approx. 9	 – approx. 265 approx. 30	 – ISO VG 150 approx. 19
Flash point	39 °C (102 °F)	–	200 °C (392 °F)	approx. 260 °C (500 °F)	> 200 °C (392 °F)
Pour point	–20 °C (-4 °F)	–	–15 °C (5 °F)	– –	< –30 °C (–22 °F)
NSF approval	Not applicable	Not applicable	Not applicable	Not applicable	H1 (No: 136858)
Available pack sizes	5 l can 180 l drum –	– 0,5 kg can –	SYSTEM 24 (LAGD / LAGE) 400 ml aerosol can 5 l can	SYSTEM 24 (LAGD / LAGE) 400 ml aerosol can 5 l can	SYSTEM 24 (LAGD / LAGE) 400 ml aerosol can 5 l can
Designation	LHRP 1/ (pack size)	LGAF 3E/0.5	LAGD 125/HMT68 LAGE 125/HMT68 LAGE 250/HMT68 LHMT 68/ (packsize)	LAGD 125/HHT26 LAGE 125/HHT26 LAGE 250/HHT26 LHHT 265/ (packsize)	LAGD 125/FHF15 LAGE 125/HFP15 LAGE 250/HFP15 LHFP 150/ (packsize)

LAOS series (page 77)

Designation	Description
LAOS 09224	Oil Safe 1,5 Litre Drum
LAOS 63571	Oil Safe 2 Litre Drum
LAOS 63595	Oil Safe 3 Litre Drum
LAOS 63618	Oil Safe 5 Litre Drum
LAOS 66251	Oil Safe 10 Litre Drum
LAOS 09644	Oil Safe Storage Lid Tan
LAOS 09651	Oil Safe Storage Lid Grey
LAOS 09934	Oil Safe Storage Lid Orange
LAOS 09941	Oil Safe Storage Lid Black
LAOS 09958	Oil Safe Storage Lid Dk Green
LAOS 09965	Oil Safe Storage Lid Green
LAOS 09972	Oil Safe Storage Lid Blue
LAOS 09989	Oil Safe Storage Lid Red
LAOS 09415	Oil Safe Storage Lid Purple
LAOS 62475	Oil Safe Storage Lid Yellow
LAOS 09668	Utility Lid Tan
LAOS 09675	Utility Lid Grey
LAOS 09866	Utility Lid Orange
LAOS 09873	Utility Lid Black
LAOS 09880	Utility Lid Dk Green
LAOS 09897	Utility Lid Green
LAOS 09903	Utility Lid Blue
LAOS 09910	Utility Lid Red
LAOS 09408	Utility Lid Purple
LAOS 62451	Utility Lid yellow
LAOS 09682	Stretch Spout Lid Tan
LAOS 09699	Stretch Spout Lid Grey
LAOS 09798	Stretch Spout Lid Orange
LAOS 09804	Stretch Spout Lid Black
LAOS 09811	Stretch Spout Lid Dk Green
LAOS 09828	Stretch Spout Lid Green
LAOS 09835	Stretch Spout Lid Blue
LAOS 09842	Stretch Spout Lid Red
LAOS 09392	Stretch Spout Lid Purple
LAOS 62437	Stretch Spout Lid Yellow

Designation	Description
LAOS 09705	Stumpy Spout Lid Tan
LAOS 09712	Stumpy Spout Lid Grey
LAOS 09729	Stumpy Spout Lid Orange
LAOS 09736	Stumpy Spout Lid Black
LAOS 09743	Stumpy Spout Lid Dk Green
LAOS 09750	Stumpy Spout Lid Green
LAOS 09767	Stumpy Spout Lid Blue
LAOS 09774	Stumpy Spout Lid Red
LAOS 09388	Stumpy Spout Lid Purple
LAOS 64936	Stumpy Spout Lid Yellow
LAOS 09057	Mini Spout Lid Tan
LAOS 09064	Mini Spout Lid Grey
LAOS 09088	Mini Spout Lid Orange
LAOS 09095	Mini Spout Lid Black
LAOS 09101	Mini Spout Lid Dk Green
LAOS 09118	Mini Spout Lid Green
LAOS 09125	Mini Spout Lid Blue
LAOS 09132	Mini Spout Lid Red
LAOS 09194	Mini Spout Lid Yellow
LAOS 09071	Mini Spout Lid Purple
LAOS 06919	Contents Label Tan
LAOS 06964	Contents Label Grey
LAOS 06940	Contents Label Orange
LAOS 06995	Contents Label Black
LAOS 06971	Contents Label Dk Green
LAOS 06957	Contents Label Green
LAOS 06988	Contents Label Blue
LAOS 06926	Contents Label Red
LAOS 06902	Contents Label Yellow
LAOS 06933	Contents Label Purple
LAOS 09422	Pump Reducer Nozzle
LAOS 67265	Stumpy Spout Hose Extension
LAOS 62499	Stretch Spout Hose Extension
LAOS 62567	Pump (to fit Oil Safe Utility Cans)
LAOS 65070	Oil Safe Dealer Sample Pack
LAOS 09217	Oil Safe Customer Sample Pack

Technical data

LAHD series (page 77)

Designation	LAHD 500 / LAHD 1000		
Boundary dimensions		**Permissible humidity**	0 – 100 %
– LAHD 500	Ø 91 mm × 290 mm high (3.6 × 11.4 in)	**Length of connecting tube**	600 mm (23.5 in)
– LAHD 1000	Ø 122 mm × 290 mm high (4.8 × 11.4 in)	**Connection thread**	G 1/2
Reservoir volume		**Tube material**	Polyurethane
– LAHD 500	500 ml (17 fl. oz. US)	**O–ring material**	NBR – 70 Shore
– LAHD 1000	1 000 ml (34 fl. oz. US)	**Gaskets**	NBR – 80 Shore 6 pieces
Container material	Polycarbonate / aluminium	**Other material**	Aluminum, Bronze, Stainless Steel
Allowed temperature range	– 20 to 125 °C (–4 to 255 °F)	**Suitable oil types**	Mineral and synthetic oils

LAGP 400 (page 78)

Designation	LAGP 400		
Maximum volume per stroke	20 cm^3 (1.2 in^3)	**Length**	360 mm (14 in)
Material	steel and polyethylene	**Weight**	0,35 kg (0.77 lb)

1077600 (page 78)

Designation	1077600		
Maximum pressure	40 MPa (5 800 psi)	**Length**	380 mm (14.9 in)
Volume/stroke	1,5 cm^3 (0.09 in^3)	**Weight**	1,5 kg (3.3 lb)

LAGH 400 (page 79)

Designation	LAGH 400		
Maximum pressure	30 MPa (4 350 psi)	**Length**	370 mm (14.6 in)
Volume/stroke	approx. 0,8 cm^3 (0.049 in^3)	**Weight**	1,5 kg (3.3 lb)

LAGG 400B (page 79)

Designation	Description
LAGG 400B	SKF Battery driven grease gun (with 230 V charger)
LAGG 400B/US	SKF Battery driven grease gun (with 110 V charger)
Maximum operating pressure	40 MPa (5 800 psi)
Min. burst pressure pump	80 MPa (11 600 psi)
Grease nozzle	4 jaws (suitable for nipples according to DIN 71412)
Operating Temperature Range	-15 to +50 °C (5 to 120 °F)
Grease NLGI	NLGI 000 to NLGI 2
Weight/dimensions:	
Dimensions of grease gun	
Including battery (L × H × D)	410 × 230 × 80 mm (16.2 × 9 × 3.2 in)
Weight of grease gun (including battery)	3,1 kg (6.8 lbs)
Dimensions of carrying case (W × D × H)	480 × 390 × 130 mm (18.9 × 15.3 × 5.1 in)
Total weight (including case)	5,4 kg (11.9 lbs)

Replacement parts

Designation	Description
LAGG 400B-1	High pressure hose 750 mm (29.5 in) with gripping nozzle
LAGG 400B-2	Battery pack

TMBA G11D (page 80)

Designation	TMBA G11D		
Pack size	50 pairs	**Colour**	blue
Size	9		

LAGF series (page 81)

Designation	LAGF 18	LAGF 50
Maximum pressure	3 MPa (430 psi)	3 MPa (430 psi)
Volume/stroke	approx. 45 cm^3 (1.5 fl oz.)	approx. 45 cm^3 (1.5 fl oz.)
Suitable drum dimensions:		
– inside diameter	265 – 285 mm (10.4 – 11.2 in)	350 – 385 mm (13.8 – 15.2 in)
– maximum inside height	420 mm (16.5 in)	675 mm (26.6 in)
Weight	5 kg (11 lb)	7 kg (15 lb)

LAGG series (page 81)

Designation	LAGG 18M	LAGG 18AE	LAGG 50AE	LAGG 180AE	LAGT 180
Description	Grease pump for 18 kg drums	Mobile grease pump for 18 kg drums	Grease pump for 50 kg drums	Grease pump for 180 kg drums	Trolley for drums up to 200 kg
Pumping	Manual	Air–pressure	Air–pressure	Air–pressure	n.a.
Max. pressure	50 MPa (7 250 psi)	42 MPa (6 090 psi)	42 MPa (6 090 psi)	42 MPa (6 090 psi)	n.a.
SKF Drum	18 kg (39.6 lb)	18 kg (39.6 lb)	50 kg (110 lb)	180 kg (396 lb)	180 kg (396 lb)
Inner diameter	265 – 285 mm (10.43 – 11.22 in)	265 – 285 mm (10.43 – 11.22 in)	350 – 385 mm (13.78 – 15.16 in)	550 – 590 mm (21.65 – 23.23 in)	n.a.
Note	Stationary	Mobile	Stationary	Stationary	Mobile
Volume/stroke	1,6 cc	–	–	–	–
Volume/min.	–	200 cc	200 cc	200 cc	–
Suitable grease NLGI class	000 – 2	0 – 2	0 – 2	0 – 2	–

LAGN 120 (page 82)

Designation	LAGN 120		
Max working pressure	40 MPa (5 800 psi)	**Standard**	DIN 71412
Min burst pressure	80 MPa (11 600 psi)	**Material**	Hardened steel

TMTP 200 (page 85)

Designation	TMTP 200		
Temperature range	–40 to 200 °C (–40 to 392 °F)	**Switch off**	Button or automatic after 5 minutes
Accuracy electronics	≤ 0,5 °C (≤ 0,9 °F)	**Display indications**	Temperature, °C or °F, maximum temperature, out of range, defective probe, low battery
Display resolution	1 °C/°F		
Probe	Integrated K–Type		
Dimensions	163 × 50 × 21 mm (6.4 × 2 × 0.8 in)	**IP rate**	IP 65
Weight	95 g (0.2 lb)	**Drop resistance**	1 m (3.2 ft)
Battery	3 × AAA (LR03)		
Average battery lifetime	4 000 hours		

TMTL 500 (page 85)

Designation	TMTL 500		
Temperature range	–60 to 500 °C (–76 to 932 °F)	**Emissivity**	Pre–set 0,95
Environmental limits	Operation 0 to 50 °C (32 to 120 °F) 10 to 95% R.H.	**Laser wavelength**	635 – 650 nm
	Storage – 20 to 65 °C (–4 to 150 °C) 10 to 95% R.H.	**Laser**	Class 2
Full range accuracy	(Tamb = 23 +/- 3 °C) +/-2% of reading or 2 °C (whichever is greater)	**Maximum laser power**	1 mW
		Dimensions	175 × 72 × 39 mm (6.9 × 2.8 × 1.5 in)
Response time	500 – 1000 msec	**Packed**	Carton box
Display	LCD	**Weight**	180 g (0,4lb)
Displayed resolution	0,1 °C/F from –9,9~199,9, otherwise 1 °C/F	**Battery**	2 × AAA Alkaline type IEC LR03
Distance to spot size	11:1	**Battery lifetime**	18 hours
Spectral response	8 – 14 µm	**Switch off**	Automatic after 15 seconds after trigger is released
User selectable backlit display	No, permanently on	**EMC standards**	EN 61326:1997+A1+A2
User selectable laser pointer	No, permanently on	**Laser standards**	CFR 1040–10 / 60825-1

TMTL 1400K (page 86)

Designation	TMTL 1400K		
Temperature range using infrared	–60 to 500 °C (–76 to 932 °F)	**Alarm modes**	High and low level alarm level with warning bleep
Temperature using probe	–64 to 1 400 °C (–83 to 1 999 °F)		
Probe supplied	TMDT 2-30, suitable for use up to	**Laser wavelength**	630 – 650 nm
	900 °C (1 650 °F)	**Laser**	Class 2
Probe types suitable	K type probes	**Maximum laser power**	1 mW
Environmental limits	Operation 0 to 50 °C (32 to 120 °F)	**Dimensions**	175 × 72 × 39 mm (6.9 × 2.8 × 1.5 in)
	10 to 95% R.H.	**Packed**	Sturdy carrying case
	Storage –20 to 65 °C (–4 to 150 °C)	**Case dimensions**	415 × 195 × 50 mm (16.3 × 7.7 × 2.0 in)
	10 to 95% R.H.	**Weight**	940 g (2.1 lb)
Full range accuracy	(Tamb = 23 +/- 3 °C) +/-2% of reading or 2 °C	**Battery**	2 × AAA Alkaline type IEC LR03
	(whichever is greater)	**Battery lifetime**	140 hours with laser and backlight off. Otherwise 18 hours
Response time	50 – 1 000 msec		
Display	LCD	**Switch off**	IR mode automatic after 60 seconds after trigger is released (60 minutes can be manually selected). Probe mode automatic after 12 minutes.
Displayed resolution	0,1 °C/F from –9,9~199,9, otherwise 1° C/F		
Distance to spot size	11:1		
Spectral response	8 – 14 µm		
Emissivity variable	0,1 – 1,0	**EMC standards**	EN 61326:1997+A1+A2
User selectable backlit display	On/off	**Laser standards**	CFR 1040-10 / 60825-1
User selectable laser pointer	On/off		
Measurement modes	Max, min, average, differential, probe/ IR dual temperature modes		

Technical data

TMTL 2400K (page 87)

Designation	**TMTL 2400K**
Description	SKF Dual Laser Infrared Thermometer
Temperature range using infrared	–60 to 1 000 °C (–76 to 1 832 °F)
Temperature using probe	–64 to 1 400 °C (–83 to 1 999 °F)
Probe supplied	TMDT 2-30, suitable for use up to 900 °C (1 650 °F)
Environmental limits	Operation 0 to 50 °C (32 to 120 °F) 10 to 95% R.H. Storage –20 to 65 °C (–4 to 150 °C) 10 to 95% R.H.
Full range accuracy	(Tamb = 23 +/– 3 °C) +/–2% of reading or 2 °C (whichever is greater)
Response time	1 sec
Display	LCD
Displayed resolution	0,1 °C/F from –9,9~199,9, otherwise 1° C/F
Distance to spot size	50:1
Spectral response	8–14 µm
Emissivity variable	0,1 – 1,0
User selectable backlit display	on/off
User selectable laser pointer	on/off
Measurement modes	Max, min, average, differential, probe/ IR dual temperature modes
Alarm modes	High and low level alarm level with warning bleep
Laser wavelength	630 – 650 nm
Laser	Class 2
Maximum laser power	1 mW
Dimensions	203,3 × 197 × 47 mm (8.0 × 7.7 × 1.8 in)
Packaging	Sturdy carrying case
Case dimensions	340 × 200 × 65 mm (13.4 × 7.9 × 2.6 in)
Weight	Total (incl. case): 705 g (1.55 lbs) TMTL 2400K: 370 g (0.815 lbs)
Battery	2 × AAA Alkaline type IEC LR03
Battery lifetime	140 hours with laser and backlight off. Otherwise 18 hours
Switch off	IR mode automatic after 60 seconds after trigger is released (60 minutes can be manually selected). Probe mode automatic after 12 minutes
EMC standards	EN 61326:1997+ A1 + A2
Laser standards	CFR 1040-10 / 60825-1

K–type thermocouple probes (page 87 – 88)

Probe type	K–type thermocouple (NiCr/NiAl) acc. IEC 584 Class 1
Accuracy	± 1,5 °C (2,7 °F) up to 375 °C (707 °F) ± 0,4% of reading above 375 °C (707 °F)
Handle	110 mm (4.3 in) long
Cable	1 000 mm (39.4 in) spiral cable (excl. TMDT 2-31, -38, -39, 41)
Plug	K–type mini–plug (1 260-K)

TMTI 2 (page 89)

Designation	**TMTI 2**
Performance	
Field of view (FOV)	20° x 15°
Focus	Manual
Minimum focus	30 cm (11,8 in)
Spectral response	8 µm to 14 µm
Thermal sensitivity	150 mK (0,15 °C) @ 25 °C scene
Temperature detector	160 x 120 pixels un–cooled microbolometer
Measurement	
Temperature range	–10 °C to + 250 °C (14 °F to 482 °F)
Radiometry	Two movable temperature measurement cursors
Temperature difference measurement	
Emissivity correction	User selectable 0,2 to 1,0 in steps of 0,01 with reflected ambient temperature compensation
Accuracy	The greater of ± 2 °C (± 3,6 °F) or 2 %
Display	3½" colour LCD with LED backlight 4 colour palettes: ironbow, rainbow, HC (high contrast) rainbow and greyscale.
Image storage	
Number	Up to 1 000 images on SD card supplied
Medium	SD card (1 GB maximum)
Laser pointer	A built in Class 2 laser is supplied to highlight the central measurement area
Imager power supply	
Battery	Lithium-ion field rechargeable, replaceable batteries
Operation time	4 hours continuous operation
AC operation	AC adaptor supplied
Mechanical	
Housing	Impact resistant plastic
Dimensions	230 mm × 120 mm × 110 mm (9 × 4.7 × 4.3 in)
Weight	0,75 kg including battery (1.6 lbs)
Mounting options	Handheld & tripod mounting
Interfaces	USB type B
Environment	
Temperature operating range	–15 °C to + 45 °C (5 °F to 113 °F)
Humidity	10 % to 90 % non-condensing
Temperature storage range	–20 °C to +70 °C (–4 °F to 158 °F)
Kit contents	TMTI 2 advanced thermal imager, rechargeable battery, 12V power supply, 128 MB SD card and case USB card reader and cable, lens cap, regional plug adapters, CD ROM (instructions for use, PC software and report writer), USB cable to connect to PC, wrist strap, carrying case

TMTI 300 (page 90)

Designation	TMTI 300
Performance	
Temperature measurement range	–10 to 300 °C (14 to 572 °F)
Field of view (FOV)	20° × 20°
Spectral response	8 to 14 µm
Sensitivity	~0,3 K @ 30 °C (@ 102,2 °F)
Displayed image	96 × 96 pixels on Pocket PC. 128 × 128 pixels on PC
Detector	16 × 16 pixel array
Frame rate	8 Hz
Range	0,7m – infinity (2,29 ft – infinity)
Image storage	Up to 1000 images per Mb of memory
Laser pointer	Class II laser
Imager power supply	
Battery	4 × AA (LR6) alkaline batteries
Operation time	Up to 8 hours
AC operation	AC adaptor (supplied)
Mechanical	
Housing	Impact resistant plastic
Dimensions	120 × 125 × 80 mm (3.72 × 4.92 × 3.1 in)
Weight	<600 g (21.16 oz) not including 'Pocket PC' and handle
Mounting	Handheld & tripod mounting
Environment	
Temp. operating range	–5 to 50 °C (23 to 122 °F)
Humidity	10 % to 90 % non condensing
Temp. storage range	–20 to 80 °C (–4 to 176 °F)
CE Mark (Europe)	EMC DIRECTIVE 89/336/EEC as outlined in harmonized norm for Emission EN 61000-6-3, EN 61000-6-4, Immunity EN 61000-6-1, EN 61000-6-2
IP	40
Laser conformance	USA 21, CFR 1040.10
Included accessories	Imager & handle Software for 'Pocket PC' & PC iPaq type synchronization cable 2m RS232 connection cable – imager to PC User manual AC power supply Tool case
Computer requirements	
Pocket PC	Compatible with most 'Pocket PC' devices running Microsoft 'Pocket PC' 2000, 2002 and 2003 RS 232 to 'Pocket PC' communication cable or CompactFlash RS 232 adaptor where applicable.
PC	IBM compatible PC with a minimum of: 32Mb RAM, 300MHz processor, MS Windows (2000 and XP), RS 232 serial port (115k Baud), 16 bit colour graphics capability

TMSP 1 (page 90)

Designation	TMSP 1
Desciption	SKF Sound Pressure Meter
Frequency range	31,5Hz to 8KHz
Measuring level range	30 to 130 dB
Display	LCD
Digital display	4 digits Resolution: 0,1 dB Display update: 0,5 s
Analogue display	50 segments bar-graph Resolution: 1 dB Display update: 100 ms
Time weighting	Fast (125 ms), Slow (1 s)
Level ranges	Lo = 30 ~ 80 dB Med = 50 ~ 100 dB Hi = 80 ~ 130 dB Auto=30~130 dB
Accuracy	±1,5 dB (ref 94dB @1KHz)
Dynamic range	50 dB
Power supply	9V Alkaline, IEC 6LR61
Power life	50 hours (with alkaline battery)
Operation temperature	0 to 40ºC (32 to 104º F)
Operation humidity	10 to 90% RH
Operation altitude	Up to 2 000 m (6 560 ft) above sea level
Storage temperature	-10 to 60ºC (14 to 140º F)
Storage humidity	10 to 75% RH
Dimensions	275 x 64 x 30 mm (10.8 x 2.5 x 1.2 in)
Case dimensions	310 x 165 x 73 mm (12.2 x 6.5 x 2.8 in)
Weight	285 g (0.76 lbs) including battery
Total weight (incl.case)	730 g (1.95 lbs)

TMRS 1 (page 92)

Designation	TMRS 1
Flash rate range	40 – 12 500 flashes per minute (FPM)
Flash rate accuracy	+/- 0,5 FPM or +/- 0,01% of reading, whichever is greater
Flash setting resolution	100 to 9999 FPM – 0,1FPM, 10 000 to 12 500 FPM -1FPM
Tachometer range	40 – 59 000 RPM
Tachometer accuracy	+/- 0,5 RPM or +/- 0,01% of reading, whichever is greater
Flash tube	Xenon, 10W, TMRS 1-BULB
Flash tube life	100 million flashes
Flash duration	9 – 15 µ sec
Light power	154 mJ per flash
Battery type	NiMH, rechargeable, removable
Battery capacity	2,6 AmpHr
Battery charge time	2 – 4 hours, using supplied AC adapter
Run time per charge	2,5 hours at 1 600 FPM, 1,25 hours at 3 200 FPM
Battery charger AC input	100 – 240 VAC, 50/60 Hz
Display	8 character by 2 line LCD, alphanumeric
Display update	continuous
Display resolution	100 to 9 999 FPM – 0,1FPM, 10 000 to 12 500 FPM – 1FPM
Time base	Crystal oscillator, 100 ppm accuracy
Controls	Power, × 2, ×1/2, phase shift, external trigger
External trigger input	0 – 5V TTL type via stereo phono jack
EXTL. trigger to flash delay	5 µ sec maximum
Clock output 0 – 5V TTL	Type signal via stereo phono jack
Colour	Grey
Housing	Impact & oil resistant polycarbonate
Weight	650 g / 1 lb, 4 oz.
Operating temperature	0 °C to 45 °C (32 °F to 113° F)
Storage temperature	–20 °C to 45 °C (–4 °F to 113 °F)

Technical data

TMRT series (page 91)

Designation	TMRT 1 / TMRT 1Ex		
Display	Inverting LCD Vertical 5 digit display	Resolution range features	Fully Auto ranging up to 0,001 digit or ± 1 digit fixed, user selectable
Display functions	180º Inverting	On target indicator	Yes
Rotational speed range	Optical mode: 3 – 99,999 rpm (or equivalent in rps) Contact mode: Max. 50 000 rpm for 10 sec (or equivalent in rps)	Low battery indicator	Yes
		Memory features	Last reading held for 1 minute Program settings retained in memory after power off
Linear speed range	0,30 – 1 500,0 Metres or Yds/min. (4 500 ft/min) or equivalent in seconds	Auto switch off	After 1 minute
Measurement modes	Optical; rpm and rps (also Count and Time) Via contact adaptor; rpm and rps, metres, yards, feet, per min and per sec. Count total revs, metres, feet, yards Measure Time interval in seconds between pulses (reciprocal rate) Speed Capture feature–Maximum, Minimum or Average rate	Remote input for laser remote sensor TMRT 1-56	Yes, TMRT 1 only
		Contact adaptor	Included complete with rpm cone and removable metric wheel assembly
		Battery type TMRT 1	4 × AAA alkaline cells
		Battery type TMRT 1 Ex	Only use 4 × Duracell “Procell” AAA cells
		Unit dimensions	213 × 40 × 39 mm (8.3 × 1.5 × 1.5 in)
		Unit weight	170 g (5.9 oz)
		Carrying case dimensions	238 × 49 × 102 mm (9.3 × 1.9 × 4.0 in)
Laser optical range	50 mm – 2 000 mm (1.9 – 78.7 in)	Total weight (incl. case)	355 g (12.5 oz)
Angle of operation	± 80º	Warranty	12 months
Light source	Class II laser diode	Intrinsically safe classification (TMRT 1Ex only)	II 2 G EEx ia IIC T4
Accuracy speed modes only	0,01%, ± 1 digit	EC Type Examination Certificate	Baseefa03ATEX0425X

Product and accessories ordering details

Designation	Description
TMRT 1	Multi function laser and contact tachometer
TMRT 1Ex	Intrinsically safe multi function laser and contact tachometer
TMRT 1-56	Laser remote sensor, for TMRT 1 only Ø 22 × 65 mm (0.8 × 2.5 in)
TMRT 1-60	Bracket for laser remote sensor

TMES series (page 93)

Designation	TMES 1	TMES 2
Description	Endoscope	Endoscope
Fibre		
Fibre material	Acrylic	Acrylic
Number of pixels	3 500	7 400
Fibre stand diameter	35 µm	13 µm
Allowable ratio of fibre breakage	2% maximum	0,5% maximum
Cord		
Fibre material	SUS304 coated with PVC	SUS304 coated with PVC
Minimum bending radius	R 40 mm	R 40 mm
Tube length	1 metre (3.3 ft)	3 metres (9.9 ft)
Tube diameter	8mm (0.3 in)	5 mm (0.2 in)
Light source		
Light type	3,5V 0,7A 2,55W	3,5V 0,7A 2,55W
Power supply	3x C (LR 14) batteries	3x C (LR 14) batteries
Optical data		
Focal direction	straight	straight
Angle of view	60°	55°
Focal length	10 mm (0.39 in) – ∞ (fixed focus)	10 mm (0.39 in) – ∞ (fixed focus)
Water resistance	Objective lens and fibre image tube is water resistant at an atmospheric pressure between 1 and 1,3 bar. Eye piece is not waterproof.	Objective lens and fibre image tube is water resistant at an atmospheric pressure between 1 and 1,3 bar. Eye piece is not waterproof.
Working temperature range	–20 °C to 60 °C (–4 °F to 140 °F)	–20 °C to 60 °C (–4 °F to 140 °F)
Weight (case and contents)	1 135 g (2.5 lbs)	1 135 g (2.5 lbs)
Measurements (case size)	360 × 270 × 80 mm (14.1 × 10.6 × 3.1 in)	360 × 270 × 80 mm (14.1 × 10.6 × 3.1 in)

TMST 3 (page 94)

Designation	TMST 3		
Description	Electronic stethoscope	Auto switch off	Yes, after 2 min
Frequency range	30 Hz-15kHz	Battery	4 × AAA/R03 (included)
Operating temperature	–10 to + 45 °C (14 – 113 °F)	Battery lifetime	30 hours (continuous use)
Output volume	Adjustable in 32 levels	Dimensions handset	220 × 40 × 40 mm (8.6 × 1.6 × 1.6 in)
Led indicator	Power on Sound volume Battery low	Probe length	70 and 220 mm (2.8 and 8.7 in)
		Weight	
		Total weight	1 560 g (3.4 lb)
Maximum recorder output	250 mV	Instrument	162 g (0.35 lb)
Headset	48 Ohm (with ear defender)	Headset	250 g (0.55 lb)

TMSU 1 (page 94)

Designation	TMSU 1		
Desciption	Ultrasonic leak detector	Battery life	Typically 20 hours
Amplification	7 levels: 20, 30, 40, 50, 60, 70 and 80 dB	Dimensions	Body: 170 × 42 × 31 mm (6.70 × 1.65 × 1.22 in) Flexible tube length: 400 mm (15.75 in)
Ultrasound sensor	Open sensor with a 16 mm (1/2 in Ø) diameter (19 mm – $\frac{3}{4}$ in Ø – exterior), central frequency of 40kHz	Weight	412 g incl. batteries (14.5 oz)
		Operating temperature range	–10 °C to 50 °C (14 °F to 122 °F)
Detected frequencies	38,4 kHz, ± 2 kHz (–3 dB)		
Power	Two alkaline AA batteries, 1,5 V. Rechargeable batteries can also be used but the usage time will be reduced		

TMEH 1 (page 95)

Designation	TMEH 1		
Suitable oil types	mineral and synthetic oils	Battery	9V Alkaline IEC 6LR61
Repeatability	better than 95%	Battery lifetime	> 150 hours or 3 000 tests
Read-out	green/red grading + numerical value (0 – 100)	Dimensions	250 × 95 × 32 mm (instrument) (9.8 × 3.7 × 1.3 in)

TMVM 1 (page 95)

Designation	TMVM 1		
Description	Handheld viscometer	Viscometer dimension (w × d × h)	175 × 88 × 170 mm (6.8 × 3.4 × 6.6 in)
Dynamic viscosity range (mPas)	30-1 300 with rotor 3 (30-400 000 using optional rotors)	Rotor dimension	D=45,1 mm (1.7 in) h=47 mm (1.8 in)
		Rotor material	Stainless steel
Motor rated voltage	4.0 VDC	Measuring cup dimension	D=52,6 mm (2.0 in) h=75 mm (2.9in)
Motor rated speed	62.5 rpm	Measuring cup material	Stainless steel
Rotor supplied	R3	Battery	4 × AA (IEC type LR06) alkaline
Repeatability	< 1 % of total range	Total weight (including case)	2,0 kg (4.4 lbs)
Accuracy	± 3 % of total range with supplied rotor R3	Packaging	Sturdy carrying case
Operating temperature	10 °C – 40 °C (50 °F – 104 °F)	Certificate of calibration	Yes
Oil sample volume	Approx. 150 ml (5.1 US fl.oz)		

CMVP series (page 96)

Designation	CMVP 40 / CMVP 50		
Vibration pick–up	Piezo electric acceleration integrated sensor (compression type)	Hold indication	HOLD
		Power	2 × CR2032 lithium batteries
Measurement range	1 to 55,0 mm/s (RMS) 0.06 to 3.00 in/s (eq. Peak)	Battery lifetime	170 mA hours current consumption Measurement mode: 7,5 mA HOLD mode 3,0 mA
Tolerance:	± 10% and 2 digits measured at 80Hz (2 digits)	Auto power off function	Power is turned off Approximately 2 minutes after last ON or HOLD operation
Frequency range	Overall vibration – 10 Hz to 1 000 Hz (tolerance measured within the frequency range are in accordance with ISO 3945 and 2 digits) acceleration enveloping – 10 kHz to 30 kHz	Dimensions	17,8 × 30,5 × 157,5 mm (0.7 × 1.2 × 6.2 in)
		Weight	approximately 77 g (2.7 oz) with batteries
Display	Measurement value: 3,5 digit LCD	Ambient operating conditions	–10 to 50 °C (14 to 122 °F) 20 to 90% relative humidity
Display cycle	Approximately 1 second		
Overload indication	OVER		
Battery replacement indication	BATT		

TMHS 75 and TMHS 100 (page 106)

Designation	TMHS 75	TMHS 100
Contents	1 × hydraulic spindle 2 × extension pieces; 50 and 100 mm (2.0 and 3.9) 1 × nosepiece	1 × hydraulic spindle 3 × extension pieces; 50, 100 and 150 mm (2.0, 3.9 and 5.9 in) 1 × nosepiece
Maximum withdrawal force	75 kN (8.4 ton US)	100 kN (11.2 ton US)
Piston stroke	75 mm (3.0 in)	80 mm (3.1 in)
Body thread	UN $1\frac{1}{4}$ × 12	UN $1\frac{1}{2}$ × 16
Nose piece diameter	35 mm (1.4 in)	30 mm (1.2 in)
Maximum reach	204 mm (8.0 in)	354 mm (13.9 in)
Weight	2,7 kg (6.0 lb)	4,5 kg (10.0 lb)

Technical data

TMMA series (page 104)

Designation	TMMA 60	TMMA 80	TMMA 120
General			
Width of grip external, minimum	36 mm (1.4 in)	52 mm (2.0 in)	75 mm (3.0 in)
Width of grip external, maximum	150 mm (5.9 in)	200 mm (7.8 in)	250 mm (9.8 in)
Effective arm length	150 mm (5.9 in)	200 mm (7.8 in)	250 mm (9.8 in)
Maximum withdrawal force	60 kN (6.7 ton US)	80 kN (9.0 ton US)	120 kN (13.5 ton US)
Total weight	4,0 kg (8.8 lb)	5,7 kg (12.6 lb)	10,6 kg (23.4 lb)
Claw dimensions			
Claw height	7,5 mm (0.30 in)	9,8 mm (0.39 in)	13,8 mm (0.54 in)
Claw length	15 mm (0.6 in)	18 mm (0.7 in)	24 mm (0.9 in)
Claw width	20 mm (0.8 in)	28 mm (1.1 in)	40 mm (1.6 in)
Force generators			
Hexagon on puller or adapter	27 mm	30 mm	32 mm
Hexagon on mechanical spindle	17 mm	22 mm	24 mm
Max torque	105 Nm (75 lbf ft)	175 Nm (125 lbf ft)	265 Nm (195 lbf ft)
Diameter nose piece	24 mm (0.9 in)	26 mm (1.0 in)	28 mm (1.1 in)
Adapter: possible to upgrade to hydraulic version	no	yes	yes
Spare parts			
Arm	TMMA 60-1	TMMA 80-1	TMMA 120-1
Spindle with nose piece (and adapter)	TMMA 60-2	TMMA 80-2	TMMA 120-2
Opening mechanism	TMMA 60-3	TMMA 75H/80-3	TMMA 100H/12-3
Accessories			
Puller protection blanket	TMMX 210	TMMX 280	TMMX 350
Gloves	TMBA G11W	TMBA G11W	TMBA G11W
Hydraulic spindle	–	TMHS 75	TMHS 100
Spindle grease	LGEV 2/0.035	LGEV 2/0.035	LGEV 2/0.035
Tri– section pulling plates	TMMS 50	TMMS 50 / TMMS 100	TMMS 50 / TMMS 100 /TMMS 160

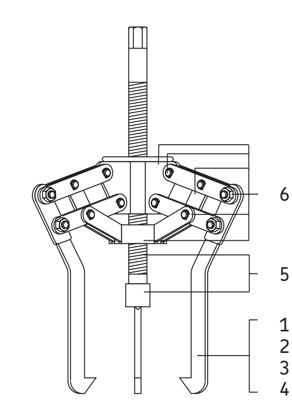
TMMA H series (page 104)

Designation	TMMA 75H	TMMA 100H
General:		
Width of grip external, minimum	52 mm (2 in)	75 mm (3 in)
Width of grip external, maximum	200 mm (7.8 in)	250 mm (9.8 in)
Effective arm length	200 mm (7.8 in)	250 mm (9.8 in)
Maximum withdrawal force	75 kN (8.4 ton US)	100 kN (11.2 ton US)
Total weight	7,2 kg (15.9 lb)	13,2 kg (29 lb)
Claw Dimensions		
Claw height	9,8 mm (0.39 in)	13,8 mm (0.54 in)
Claw length	18 mm (0.7 in)	24 mm (0.9 in)
Claw width	28 mm (1.1 in)	40 mm (1.6 in)
Force generator		
Hydraulic spindle	TMHS 75	TMHS 100
Piston stroke	75 mm (3.0 in)	80 mm (3.1 in)
Body thread	UN 1,25" × 12	UN 1.5" × 16
Diameter nose piece	35 mm (1.4 in)	30 mm (1.2 in)
Spare parts		
Arm	TMMA 75H-1	TMMA 100H-1
Opening mechanism	TMMA 75H/80-3	TMMA 100H/12-3
Hydraulics extension piece set	TMHS 5T	TMHS 8T
Accessories		
Hydraulic spindle	TMHS 75 (included)	TMHS 100 (included)
Puller protection blanket	TMMX 280	TMMX 350
Gloves	TMBA G11W	TMBA G11W
Tri section pulling plates	TMMS 50 TMMS 100	TMMS 50 TMMS 100 TMMS 160

TMMP series (page 108)

Designation	No. of arms	Width of grip		Effective length of arms*		Maximum withdrawal force		Weight	
	qty	mm	in	mm	in	kN	ton US	kg	lb
TMMP 6	3	50 – 127	2.0 – 5.0	120	4.7	60	6.7	4,0	8.8
TMMP 10	3	100 – 223	3.9 – 8.7	207	8.2	100	11.2	8,5	19
TMMP 15	3	140 – 326	5.5 – 12.8	340	13.4	150	17.0	21,5	46

** other arm lengths available according to part ordering details*



Part ordering details

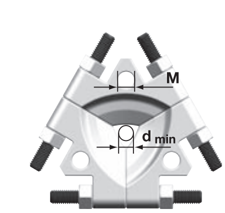
No.	Designation	Description	TMMP 6		TMMP 10		TMMP 15	
			mm	in	mm	in	mm	in
1	**TMMP ...-1**	Arm–length	120	4.7	207	8.2	260	10.2
2	**TMMP ...-2**	Arm–length	220	8.6	350	13.8	340	13.4
3	**TMMP ...-3**	Arm–length	370	14.5	460	18.1	435	17.1
4	**TMMP ...-4**	Arm–length	470	18.5	710	27.9	685	27.0
5	**TMMP ...-5**	Spindle with centre nib						
6	**TMMP ...-1K**	Stand, boss and complete set of pins, bolts and link arms (per arm)						

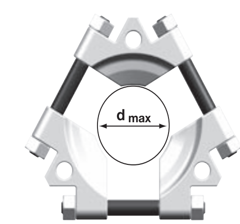
TMMA 100H/SET (page 105)

Designation	TMMA 100H/SET		
General		**Max diameter**	350 mm (13.8 in)
Width of grip external, minimum	75 mm (3 in)	Length	1200 mm (47 in)
Width of grip external, maximum	250 mm (9.8 in)	Width	580 mm (19 in)
Effective arm length	250 mm (9.8 in)	Weight	0,6 kg (1.4 lb)
Maximum withdrawal force	100 kN (11.2 ton US)	**Case**	
Claw dimensions		Height	270 mm (11 in)
Claw height	13,8 mm (0.54 in)	Length	680 mm (27 in)
Claw length	24 mm (0.9 in)	Width	320 mm (13 in)
Claw width	40 mm (1.6 in)	Weight	12,0 kg (26.5 lb)
Force generator		**Spare parts**	
Hydraulic spindle	TMHS 100	Arm	TMMA 100H-1
Piston stroke	80 mm (3.1 in)	Opening mechanism	TMMA 100H/12-3
Body thread	UN 1,5"x16	Hydraulic extension piece set	TMHS 8T
Diameter nose piece	30 mm (1.2 in)	**Accessories**	
Tri-section pulling plate	TMMS 160	Puller protection blanket	TMMX 350 (included)
Width of grip shaft, minimum	50 mm (2.0 in)	Hydraulic spindle	TMHS 100 (included)
Width of grip shaft, maximum	160 mm (6.3 in)	Tri section pulling plates	TMMS 160 (included)
Weight	5,9 kg (13.0 lb)	Gloves	TMBA G11W
Puller protection blanket	TMMX 350		

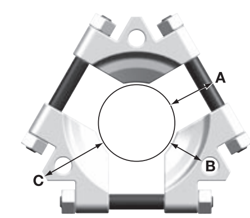
TMMS series (page 105)

Designation	Width of grip				
	d min		d max		M
	mm	in	mm	in	mm
TMMS 50	12	0.5	50	2.0	–
TMMS 100	26	1.0	100	3.9	M16 × 2
TMMS 160	50	2.0	160	6.3	M16 × 2
TMMS 260	90	3.6	260	10.2	M22 × 2.5
TMMS 380	140	5.5	380	15.0	M32 × 2.5





Designation	A		B		C		Maximum Withdrawal Force (F max)		Weight	
	mm	in	mm	in	mm	in	kN	ton US	kg	lb
TMMS 50	20	0.8	–	–	32	1.3	80	9	0,5	1.1
TMMS 100	36	1.4	34	1.4	60	2.4	200	23	2,6	5,7
TMMS 160	45	1.8	52	2.1	82	3.3	300	34	5,9	13
TMMS 260	70	2.8	81	3.2	110	4.3	450	51	18,4	41
TMMS 380	81	3.2	97	3.8	138	5.4	600	68	50,3	110



TMMP series (page 107)

Designation	No. of arms	Width of grip		Effective length of arms		Maximum withdrawal force		Weight	
	qty	mm	in	mm	in	kN	ton US	kg	lb
TMMP 2x65	2	15 – 65	0.6 – 2.6	60	2.4	6,0	0.7	0,5	1.2
TMMP 2x170	2	25 – 170	1.0 – 6.7	135	5.3	18,0	2.0	2,1	4.7
TMMP 3x185	3	40 – 185	1.6 – 7.3	135	5.3	24,0	2.7	2,9	6.4
TMMP 3x230	3	40 – 230	1.6 – 9.1	210	8.3	34,0	3.8	5,8	13
TMMP 3x300	3	45 – 300	1.8 – 11.8	240	9.4	50,0	5.6	8,6	19

TMMR F series (page 107)

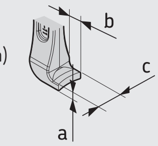
Designation	Width of grip external pull		Width of grip internal pull		Effective arm length		Maximum withdrawal force		Weight	
	mm	in	mm	in	mm	in	kN	ton US	kg	lb
TMMR 40F	23 – 48	0.9 – 1.9	59 – 67	2.3 – 2.6	65	2.6	15	1.7	0,3	0.8
TMMR 60F	23 – 68	0.9 – 2.7	62 – 87	2.4 – 3.4	80	3.2	15	1.7	0,4	0.8
TMMR 80F	41 – 83	1.6 – 3.3	93 – 97	3.7 – 3.8	94	3.7	30	3.4	1,0	2.2
TMMR 120F	41 – 124	1.6 – 4.8	93 – 138	3.7 – 5.4	120	4.7	30	3.4	1,2	2.6
TMMR 160F	68 – 164	2.7 – 6.5	114 – 162	4.5 – 6.4	130	5.1	40	4.5	2,3	5.2
TMMR 200F	67 – 204	2.6 – 8.0	114 – 204	4.,5 – 8.0	155	6.1	40	4.5	2,6	5.8
TMMR 250F	74 – 254	2.9 – 10.0	132 – 252	5.2 – 9.9	178	7.0	50	5.6	4,4	9.7
TMMR 350F	74 – 354	2.9 – 14.0	135 – 352	5.3 – 13.8	233	9.2	50	5.6	5,2	11.5
TMMR 8	Complete kit of 8 pullers on a counter stand									

Technical data

TMHP 10E (page 109)

Designation	TMHP 10E
Description	Advanced hydraulic jaw puller kit
Contents	1 × arm–assembly stand 3 × arms, 120 mm (4.7 in) 3 × arms, 170 mm (6.7 in) 3 × arms, 200 mm (7.8 in) 1 × hydraulic spindle TMHS 100 3 × extension pieces for hydraulic spindle; 50, 100, 150 mm (2, 4, 6 in) 1 × nosepiece with centre point for hydraulic spindle
Maximum stroke	80 mm (3.1 in)
Weight complete kit	14,5 kg (32 lb)
Cycle life hydraulic cylinder	Minimum 5 000 cycles up to 100 kN (11.2 US ton force)
Threading hydraulic cylinder	UN 1$\frac{1}{2}$" × 16 tpi
Safety valve setting hydraulic cylinder	105 kN (11.8 US ton force)
Carrying case dimensions	578 × 410 × 70 mm (23 × 16 × 2.8 in)
Nominal working force	100 kN (11.2 US ton force)

Arm set 1 (3 × TMHP 10E-10)	
Effective arms length	120 mm (4.7 in)
Width of grip	75 – 170 mm (3,0 – 6.7 in)
Claw dimensions	a = 6 mm (0.2 in) b = 15 mm (0.6 in) c = 25 mm (1 in)
Arm set 2 (3 × TMHP 10E-11)	
Effective arms length	170 mm (6.7 in)
Width of grip	80 – 250 mm (3.1 – 9.8 in)
Claw dimensions	a = 6 mm (0.2 in) b = 12 mm (0.5 in) c = 25 mm (1 in)
Arm set 3 (3 × TMHP 10E-12)	
Effective arms length	200 mm (7.8 in)
Width of grip	110 – 280 mm (4.3 – 11 in)
Claw dimensions	a = 6 mm (0.2 In) b = 12 mm (0.5 in) c = 25 mm (1 in)



Part ordering details

Designation	Description	Designation	Description
TMHS 100	Advanced hydraulic spindle, 100 kN	TMHP 10E-10	120 mm arm (4.7 in)
TMHS 8T	Set of extension pieces an nose piece for the hydraulic spindle	TMHP 10E-11	170 mm arm (6.7 in)
TMHP 10E-5	Arm–assembly stand, centre, bolts and nuts	TMHP 10E-12	200 mm arm (7.8 in)

TMHP series (page 108)

Designation*	No. of arms	Width of grip		Effective length of arms		Stroke		Maximum working pressure		Maximum withdrawal force		Weight	
	qty	mm	in	mm	in	mm	in	MPa	psi	kN	ton US	kg	lb
TMHP 15/260	3	195–386	7.7–15.2	264	10.4	100	3.9	80	11 600	150	16.9	34	75
TMHP 30/170	3	290–500	11.4–19.7	170	6.7	50	2.0	80	11 600	300	33.7	45	99
TMHP 30/350	3	290–500	11.4–19.7	350	13.7	50	2.0	80	11 600	300	33.7	47	104
TMHP 30/600	3	290–500	11.4–19.7	600	23.6	50	2.0	80	11 600	300	33.7	56	123
TMHP 50/140	3	310–506	12.2–19.9	140	5.5	40	1.6	80	11 600	500	56.2	47	104
TMHP 50/320	3	310–506	12.2–19.9	320	12.6	40	1.6	80	11 600	500	56.2	54	119
TMHP 50/570	3	310–506	12.2–19.9	570	22.4	40	1.6	80	11 600	500	56.2	56	123

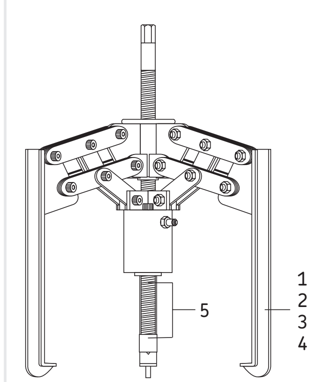
**Also available without hydraulic pump TMJL 100. Please add suffix 'X' to designation when ordering (e.g. TMHP 30/170X)*

TMHP series (page 108)

Designation	Hydraulic pump TMJL 100
Maximum pressure	100 MPa (14 500 psi)
Volume/stroke	1 cm³ (0.06 in³)
Oil container capacity	800 cm³ (48 in³)
Pressure hose	3 000 mm (118.1 in) long with quick connection coupling and nipple G 1/4 internal/external thread
Weight with gauge	13 kg (29 lb)
Oil type	filled with SKF LHMF 300

Part ordering details

No.	Designation	Description	TMHP 15		TMHP 30		TMHP 50	
			mm	in	mm	in	mm	in
1	TMHP ...-1	Arm–length	264	10.4	170	6.7	140	5.5
2	TMHP ...-2	Arm–length	344	14.2	350	13.7	320	12.6
3	TMHP ...-3	Arm–length	439	17.3	600	23.6	570	22.4
4	TMHP ...-4	Arm–length	689	27.1				
5	TMHP ...-5	Spindle with centre nib						
	TMHP ...-11	Repair kit for hydraulic cylinder						



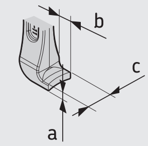
TMMX series (page 113)

Designation	Recommended maximum diameter		Length		Width		Length of strap		No. of strap	Buckle size
	mm	in	mm	in	mm	in	mm	in		in
TMMX 210	210	8.3	750	29.5	420	16.5	500	19.7	3	1
TMMX 280	280	11.0	970	38.2	480	18.9	520	20.5	3	1
TMMX 350	350	13.8	1 200	47.2	580	22.8	770	30.3	3	1 1/2

TMHC 110E (page 109)

Designation	TMHC 110E
Description	Advanced hydraulic puller kit
Contents	1 × arm–assembly stand 3 × arms, 70 mm (2.7 in) 3 × arms, 120 mm (4.7 in) 1 × separator set 1 × beam 2 × main rods 2 × extension rods, 125 mm (4.9 in) 1 × hydraulic spindle TMHS 100 2 × extension pieces for hydraulic spindle; 50, 100 mm (2.0, 3.9 in) 1 × nosepiece with centre point for hydraulic spindle
Maximum stroke	80 mm (3.1 in)
Nominal working force	100 kN (11.2 US ton force)
Cycle life hydraulic cylinder	Minimum 5 000 cycles up to 100 kN (11.2 US ton force)
Threading hydraulic cylinder	UN 1½'' × 16 tpi
Safety valve setting hydraulic cylinder	105 kN (11.8 US ton force)
Carrying case dimensions	580 × 410 × 70 mm (23 × 16 × 2.8 in)
Weight	13,5 kg (29.8 lb)

Jaw puller:	
Arms set 1 (TMHP 10E-9)	
Effective arms length	70 mm (2.7 in)
Width of grip, arms set 1	50 – 110 mm (2 – 4.3 in)
Claw dimensions	a = 5 mm (0.2 in) b = 15 mm (0.6 in) c = 25 mm (1 in)
Arms set 2 (TMHP 10E-10)	
Effective arms length	120 mm (4.7 in)
Width of grip, arms set 2	75 – 170 mm (3.0 – 6.7 in)
Claw dimensions	a = 6 mm (0.2 in) b = 15 mm (0.6 in) c = 25 mm (1 in)
Strong back puller:	
Maximum reach	255 mm (10 in)
Shaft diameter range	20 – 100 mm (0.8 – 4 in)



Part ordering details

Designation	Description
TMHP 10E-5	Arm–assembly stand, centre, bolts and nuts
TMHP 10E-9	70 mm (2.7 in) arm
TMHP 10E-10	120 mm (4.7 in) arm
TMBS 100E-1	Beam
TMBS 100E-2	Main rods, washers and nuts
TMBS 100E-3	Extension rods (2 pcs) 125 mm (4.9 in)
TMBS 100E-5	Separator set, bolts and nuts (100 mm / 4 in)
TMHS 100	Advanced hydraulic spindle, 100 kN
TMHS 8T	Set of extension pieces and a nose piece for the hydraulic spindle

TMBS E series (page 110)

Designation	TMBS 100E
Description	Advanced hydraulic strong back puller
Contents	1 × separator set 2 × main rods 2 × extension rods, 125 mm (4.9 in) 4 × extension rods, 285 mm (11.2 in) 1 × beam 1 × hydraulic spindle TMHS 100 2 × extension pieces for hydraulic spindle; 50, 100 mm (2.0, 3.9 in) 1 × nosepiece with centre point for hydraulic spindle
Maximum stroke	80 mm (3.1 in)
Nominal working force	100 kN (11.2 US ton force)
Maximum reach	816 mm (31.1 in)
Shaft diameter range	20 – 100 mm (0.8 – 4 in)
Cycle life hydraulic cylinder	Minimum 5 000 cycles up to 100 kN (11.2 US ton force)
Threading hydraulic cylinder	UN 1½'' × 16 tpi
Safety valve setting hydraulic cylinder	105 kN (11.8 US ton force)
Carrying case dimensions	580 × 410 × 70 mm (23 × 16 × 2.8 in)
Weight	13,5 kg (29.8 lb)

Designation	TMBS 150E
Description	Advanced hydraulic strong back puller
Contents	1 × separator set 2 × main rods 2 × extension rods, 125 mm (4.9 in) 4 × extension rods, 285 mm (11.2 in) 1 × beam 1 × hydraulic spindle TMHS 100 2 × extension pieces for hydraulic spindle; 50, 100 mm (2.0, 3.9 in) 1 × nosepiece with centre point for hydraulic spindle
Maximum stroke	80 mm (3.1 in)
Nominal working force	100 kN (11.2 US ton force)
Maximum reach	816 mm (31.1 in)
Shaft diameter range	35 – 150 mm (1.4 – 6 in)
Cycle life hydraulic cylinder	Minimum 5 000 cycles up to 100 kN (11.2 US ton force)
Threading hydraulic cylinder	UN 1½'' × 16 tpi
Safety valve setting hydraulic cylinder	105 kN (11.8 US ton force)
Carrying case dimensions	580 × 410 × 70 mm (23 × 16 × 2.8 in)
Weight	17 kg (37.5 lb)

Part ordering details

Designation		Description
TMHS 100	TMHS 100	Advanced hydraulic spindle, 100 kN
TMHS 8T	TMHS 8T	Set of extension pieces for the hydraulic spindle, nose piece
TMBS 100E-1	TMBS 150E-1	Beam
TMBS 100E-2	TMBS 100E-2	Main rods, nuts, washers (set)
TMBS 100E-3	TMBS 100E-3	Extension rods (2 pcs) 125 mm (4.9 in)
TMBS 100E-4	TMBS 100E-4	Extension rods (4 pcs) 285 mm (11.2 in)
TMBS 100E-5	TMBS 150E-5	Separator (complete)

TMBR series (page 115)

Designation	TMBR Bearing designation; (e.g. TMBR NU216E)
Material	Aluminium
Maximum temperature	300 °C (572 °F)

Technical data

TMBS 50E (page 110)

Designation	TMBS 50E		
Description	Mechanical strong back puller	Maximum reach	110 mm (4.3 in)
Contents	1 × separator set	Shaft diameter range	7 – 50 mm (0.3 – 2 in)
	1 × mechanical spindle	Maximum torque (T)	70 Nm (50 lbf ft)
	1 × beam	Spindle Hexagon head (AF)	19 mm (0.8 in)
	2 × main rods	Carrying case dimensions	295 × 190 × 55 mm (11.6 × 7.5 × 2 in)
Nominal working force	30 kN (3.4 US ton force)	Weight	1,8 kg (4 lb)

Part ordering details

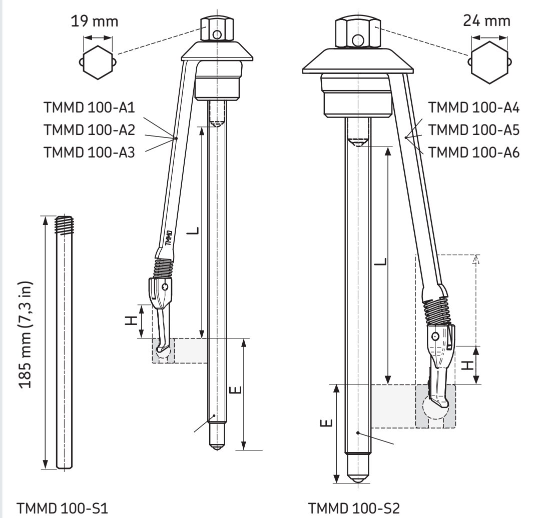
Designation	Description
TMBS 50E-1	Beam
TMBS 50E-2	Spindle
TMBS 50E-1K	Main rods, washers (4 pcs), bolts and nuts (2pcs)

TMMD 100 (page 111)

Designation	L		H		E	
	mm	in	mm	in	mm	in
TMMD 100-A1	135	5.3	16	0.6	79	3.1
TMMD 100-A2	135	5.3	16	0.6	79	3.1
TMMD 100-A3	137	5.4	23	0.9	77	3.0
TMMD 100-A4	162	6.4	26	1.0	52	2.0
TMMD 100-A5	167	6.6	> 52	> 2.0	49	1.9
TMMD 100-A6	170	6.7	> 100	> 3.9	49	1.9

Ordering details

Designation	TMMD 100
Description	Deep groove ball bearing puller kit
Kit contents	3 × puller arm TMMD 100-A1 3 × puller arm TMMD 100-A2 3 × puller arm TMMD 100-A3 3 × puller arm TMMD 100-A4 3 × puller arm TMMD 100-A5 3 × puller arm TMMD 100-A6 1 × small spindle and nut TMMD 100-S1 1 × big spindle and nut TMMD 100-S2 1 × handle
Dimensions of case	395 × 300 × 105 mm (15.5 × 11.8 × 4.1 in)
Weight	3,8 kg (8.4 lb)

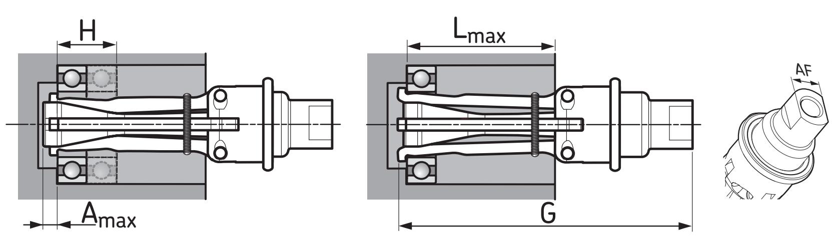


TMIP series (page 112)

Designation	TMIP 7-28	TMIP 30-60
Description	Internal bearing puller	Internal bearing puller
Weight (total kit)	3,3 kg (7.3 lb)	5,4 kg (11.9 lb)
Total sliding hammer length	412 mm (16.2 in)	557 mm (21.9 in)
Case size (w × d × h)	395 × 300 × 105 mm (15.5 × 11.8 × 4.1 in)	395 × 300 × 105 mm (15.5 × 11.8 × 4.1 in)
Spare parts	TMIP E7-9, TMIP E10-12, TMIP E15-17	TMIP E3 and TMIP E4

Technical data extractors

Extractor size	Amax Space behind bearing		Lmax Housing depth		H Bearing width		G Total extractor length		AF Adapter size
	mm	in	mm	in	mm	in	mm	in	mm
TMIP E7-9	6	0.24	38	1.5	>8	>0.3	88	3.5	15
TMIP E10-12	6	0.24	41	1.6	>17	>0.7	88	3.5	15
TMIP E15-17	6,5	0.26	49	1.9	>22	>0.9	118	4.6	15
TMIP E20-28	10	0.4	60	2.4	>24	>0.95	127	5.0	15
TMIP E30-40	11,5	0.45	97	3.8	>35	>1.4	143	5.6	19
TMIP E45-60	15	0.6	102	4.0	>64	>2.5	152	6.0	19



Kit contents TMIP series (page 112)

TMIP 7-28	TMIP 30-60
Slide hammer Extractor 7 - 9 mm (0.28 - 0.35 in) Extractor 10 - 12 mm (0.39 - 0.47 in) Extractor 15 - 17 mm (0.59 - 0.67 in) Extractor 20 - 28 mm (0.79 - 1.1 in) IFU card Carrying case	Slide hammer Extractor 30 - 40 mm (1.2 - 1.6 in) Extractor 45 - 60 mm (1.8 - 2.4 in) IFU card Carrying case

TMBP 20E (page 113)

Designation	TMBP 20E		
Effective arm length	183 mm (7.2 in)	**Set contents**	Adapters size A to F (2 pcs each) 2 x main rods (with nut support rings and nuts) 4 x extensi on rods Spindle Spindle nose piece Beam IFU card Case with inlay
Extension piece length	200 mm (7.9 in)		
Maximum arm length (incl extension pieces)	583 mm (23.0 in) for adapter F		
Maximum pulling force	55 kN (6.2 ton US)		
Maximum torque	155 Nm (114 lbf ft)		
Spindle head AF size	22 mm		
Dimensions of case	395 × 300 × 105 mm (15.5 × 11.8 × 4.1 in)		
Weight	7,5 kg (16.5 lb)	**Spare parts**	
		TMBP 20E-1	Spindle with nose piece
		Accessories	
		TMBA G11W	Gloves
		LEGV 2/0.0035	Spindle grease
		TMMX 280	Puller protection blanket

Selection chart TMBP 20E

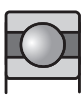
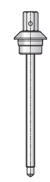
Bearing adapter	A	B	C	D	E	F
Spanner size for mounting	9 mm	11 mm	14 mm	15 mm	17 mm	19 mm
Ball adapter size	16 mm	19 mm	23,5 mm	26,5 mm	28 mm	30 mm
Bearing series						
60..	6021 6022 6024	6026 6028 6030	6032			
62..	6213 6214 6215 6216	6217 6218	6219 6220	6221	6222 6224 6226 6228 6230 6232	
63..	6309	6310 6311 6312	6313 6314 6321	6315 6316	6317 6318	6319 6320
64..	6406	6407 6408 6409	6410	6411 6412	6413	6414 6415 6416 6417 6418
160..		16026 16028 16030 16032				

Technical data

TMMK 10-35 (page 114)

Designation	TMMK 10-35		
Description	SKF Combi kit	Effective puller arm length (L)	A1- 135 mm (5.3 in)
Number of impact rings	24 (12 size A and 12 size B)		A2- 135 mm (5.3 in)
Number of sleeves	2 (size A and size B)		A3- 137 mm (5.4 in)
Impact rings bore diameter	10 – 35 mm (0.39 – 2.1 in)		A4- 162 mm (6.4 in)
Impact rings outer diameter	26 – 80 mm (1.0 – 4.7 in)		A5- 167 mm (6.6 in)
Impact sleeves bore diameter	18 and 37 mm (0.7,and 1.4 in)	Spindle hexagon heads (AF)	19 mm and 24 mm
Dead-blow hammer	TMFT 36-H, weight 1,0 kg (2.2 lb)	Sliding hammer displacement	182 mm (7.16 in)
Shaft support rings (diameter)	10, 12, 15, 17, 20, 22, 25, 28, 30 and 35 mm	Sliding hammer weight	1,0 kg (2.2 lb)
		Dimensions of the case	525 × 420 × 130 mm (20.7 × 16.5 × 5.1 in)
		Kit weight, including carrying case	7,6 kg (16,8 lbs)

Selection table TMMK 10-35

	Mounting		Dismounting		
					
DGBB	Impact Ring	Impact Sleeve	Spindle size	Puller Arm	Support ring
6000	A10-26	A	small	TMMD 100-A1	Size 1
6200	A10-30	A	small	TMMD 100-A1	
16100	A10-26	A	small	TMMD 100-A1	
6300	A10-35	A	small	TMMD 100-A2	
6001	A12-28	A	small	TMMD 100-A1	
16101	A12-28	A	small	TMMD 100-A1	
6201	A12-32	A	small	TMMD 100-A2	
6301	A12-37	A	small	TMMD 100-A3	
6002	A15-32	A	small	TMMD 100-A1	Size 2
16002	A15-32	A	small	TMMD 100-A1	
6202	A15-35	A	small	TMMD 100-A2	
6302	A15-42	A	small	TMMD 100-A3	
6003	A17-35	A	small	TMMD 100-A1	
16003	A17-35	A	small	TMMD 100-A1	
6203	A17-40	A	small	TMMD 100-A2	
6303	A17-47	A	large	TMMD 100-A4	
6403	B25-62	B	large	TMMD 100-A5	
6004	B20-42	B	small	TMMD 100-A2	Size 3
6204	B20-47	B	small	TMMD 100-A3	
6304	B20-52	B	large	TMMD 100-A4	
62/22	B25-47	B	small	TMMD 100-A3	
63/22	B25-52	B	large	TMMD 100-A4	
6005	B25-47	B	small	TMMD 100-A2	Size 4
6205	B25-52	B	small	TMMD 100-A3	
6305	B25-62	B	large	TMMD 100-A5	
62/28	B30-55	B	large	TMMD 100-A4	Size 5
63/28	B30-62	B	large	TMMD 100-A5	
6006	B30-55	B	small	TMMD 100-A2	
6206	B30-62	B	large	TMMD 100-A4	
6306	B30-72	B	large	TMMD 100-A5	
6007	B35-62	B	small	TMMD 100-A3	Size 6
6207	B35-72	B	large	TMMD 100-A5	
6307	B35-80	B	large	TMMD 100-A5	

Voltage classification EAZ series (page 114)

Each heater is available in three different voltage versions as follows:

LV	Low	190 to 230V
MV	Medium	400 to 480V
HV	High	500 to 575V

Please add the corresponding class as a suffix to the designation when ordering (e.g. EAZ 166 HV).

Control cabinets EAZ series (page 114)

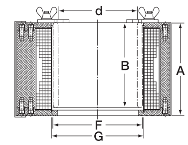
Designation		Designation	
SS 250A	230V, 50Hz, 250A	SS 250B	400V, 50Hz, 250A
SS 250C	460V, 60Hz, 250A	SS 350A	230V, 50Hz, 350A
SS 350B	400V, 50Hz, 350A	SS 350C	460V, 60Hz, 350A

A special control cabinet suitable for handling two heaters at the same time is also available.

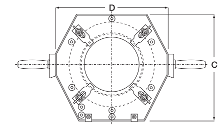
Designation		Designation	
SSD 350A	230V, 50Hz, 350A (2×)	SSD 350B	400V, 50Hz, 350A (2×)
SSD 350C	460V, 60Hz, 350A (2×)		

EAZ series (page 114)

Dimensions ring



Dimensions heater

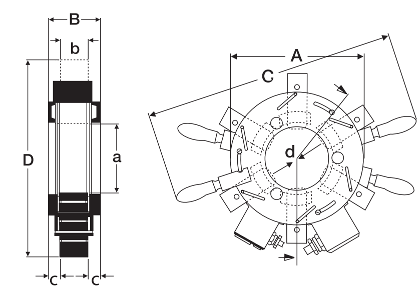


Heater designation	Voltage class	Bearing designation	Coil Current consumption	Connecting cable	Control cabinet	Dimensions ring d	B	F	G	Dimensions heater C	D	A
			A			mm	mm	mm	mm	mm	mm	mm
EAZ 166	LV MV HV	314625	170 100 75	A07 RN - F 3 × 25 A07 RN - F 3 × 16 A07 RN - F 3 × 16	SS 250	145 p6	155	166	169	350	370	176
EAZ 169	LV MV HV	313924 A	170 100 75	A07 RN - F 3 × 25 A07 RN - F 3 × 16 A07 RN - F 3 × 16	SS 250	145 p6	156	169	172	355	378	176
EAZ 174	LV MV HV	313891 A	165 95 75	A07 RN - F 3 × 25 A07 RN - F 3 × 16 A07 RN - F 3 × 16	SS 250	150 p6	156	174	177	360	388	176
EAZ 179	LV MV HV	315189 A	180 105 80	A07 RN - F 3 × 35 A07 RN - F 3 × 16 A07 RN - F 3 × 16	SS 250	160 p6	168	179	182	355	378	184
EAZ 180	LV MV HV	314190	150 85 65	A07 RN - F 3 × 25 A07 RN - F 3 × 16 A07 RN - F 3 × 16	SS 250	160 p6	130	180	183	365	390	151
EAZ 181	LV MV HV	315642/ VJ202	180 105 80	A07 RN - F 3 × 35 A07 RN - F 3 × 16 A07 RN - F 3 × 16	SS 250	165,1 p6	165	181	184	355	378	190
EAZ 190	LV MV HV	BC4B 635122	140 80 60	A07 RN - F 3 × 25 A07 RN - F 3 × 16 A07 RN - F 3 × 16	SS 250	170 p6	130	190	193	375	402	151
EAZ 202	LV MV HV	313812	165 95 70	A07 RN - F 3 × 25 A07 RN - F 3 × 16 A07 RN - F 3 × 16	SS 250	180 p6	168	202	205	375	402	190
EAZ 212	LV MV HV	314199 B	200 115 90	A07 RN - F 3 × 35 A07 RN - F 3 × 25 A07 RN - F 3 × 16	SS 250	190 p6	200	212	215	385	412	217
EAZ 222-1	LV MV HV	314553	190 110 85	A07 RN - F 3 × 35 A07 RN - F 3 × 16 A07 RN - F 3 × 16	SS 250	200 p6	170	222	225	385	412	190
EAZ 222-2	LV MV HV	313893	215 125 95	A07 RN - F 3 × 25 A07 RN - F 3 × 16 A07 RN - F 3 × 16	SS 250	200 p6	200	222	225	395	422	217
EAZ 226	LV MV HV	313811	210 120 95	A07 RN - F 3 × 35 A07 RN - F 3 × 25 A07 RN - F 3 × 16	SS 250	200 p6	192	226	229	400	425	213
EAZ 244	LV MV HV	313894 B	300 175 130	A07 RN - F 3 × 50 A07 RN - F 3 × 35 A07 RN - F 3 × 25	SS 350 SS 250	220 r6	225	244	247	410	435	247
EAZ 246	LV MV HV	313839	260 150 115	A07 RN - F 3 × 50 A07 RN - F 3 × 25 A07 RN - F 3 × 25	SS 350 SS 250	220 r6	192	246	249	410	435	214
EAZ 260	LV MV HV	313824	275 160 120	A07 RN - F 3 × 50 A07 RN - F 3 × 25 A07 RN - F 3 × 35	SS 350 SS 250	230 r6	206	260	263	425	450	227
EAZ 265	LV MV HV	635194	240 140 105	A07 RN - F 3 × 35 A07 RN - F 3 × 25 A07 RN - F 3 × 16	SS 250	240 r6	180	265	268	430	457	201
EAZ 270	LV MV HV	313921	265 155 115	A07 RN - F 3 × 50 A07 RN - F 3 × 25 A07 RN - F 3 × 25	SS 350 SS 250	240 r6	220	270	273	435	460	233
EAZ 292	LV MV HV	313823	295 170 130	A07 RN - F 3 × 50 A07 RN - F 3 × 25 A07 RN - F 3 × 25	SS 350 SS 250	260 r6	220	292	295	445	470	240
EAZ 308	LV MV HV	314719 C	335 195 145	A07 RN - F 3 × 50 A07 RN - F 3 x 35 A07 RN - F 3 x 25	SS 350 SS 250	280 r6	275	308	311	460	490	296
EAZ 312	LV MV HV	313822	285 165 125	A07 RN - F 3 × 50 A07 RN - F 3 × 25 A07 RN - F 3 × 25	SS 350 SS 250 SS 250	280 r6	220	312	315	465	490	238
EAZ 332	LV MV HV	314484 D	365 210 160	A07 RN - F 3 × 70 A07 RN - F 3 × 35 A07 RN - F 3 × 25	SS 350 SS 250	300 r6	300	332	335	480	500	322
EAZ 378	LV MV HV	314485 A	375 240 205	A07 RN - F 3 × 70 A07 RN - F 3 × 50 A07 RN - F 3 × 35	SS 350 SS 250	340 r6	350	378	381	525	555	368

Technical data

EAZ series (page 114)

Designation		EAZ 80/130		EAZ 130/170	
Connection cable	Length	5 m	16 ft	5 m	16 ft
Dimensions	A	285 mm	11.2 in	335 mm	13.2 in
	B	115 mm	4.5 in	120 mm	4.7 in
	C	555 mm	21.8 in	630 mm	24.8 in
	D	305 ... 360 mm	12.0 ... 14.1 in	335 ... 380 mm	13.2 ... 15.0 in
	a	134 mm	5.3 in	180 mm	7.1 in
	b	50 mm	2.0 in	50 mm	2.0 in
	c	35 mm	1.4 in	30 mm	1.2 in
	d	80 ... 132 mm	3.1... 5.2 in	130 ... 172 mm	5.1 ... 6.8 in
Weight		28 kg	62 lb	35 kg	77 lb



Part ordering details

Designation	Powersupply	Current	Designation	Powersupply	Current
EAZ 80/130A	2 × 230V/50Hz	40 A	**EAZ 130/170D**	3 × 230V/50Hz	43 A
EAZ 80/130B	2 × 400V/50Hz	45 A	**EAZ 130/170E**	3 × 400V/50Hz	35 A
EAZ 80/130C	2 × 460V/60Hz	25 A	**EAZ 130/170F**	3 × 460V/60Hz	23 A
EAZ 80/130D	2 × 415V/50Hz	35 A	**EAZ 130/170G**	3 × 420V/60Hz	30 A
EAZ 130/170A	2 × 230V/50Hz	60 A	**EAZ 130/170H**	3 × 415V/50Hz	30 A
EAZ 130/170B	2 × 400V/50Hz	45 A			

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